Terminal Value: What Death Was Doing for the Economy, and What Happens When We Stop It

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Chapter 1

The Boundary Condition

There is a type of financial arrangement that works perfectly, forever, so long as nobody dies.

It is called a Ponzi scheme.

The mechanics are familiar. You take money from new investors and pay it to old ones, and you call the payment a return. Nothing is produced. Nothing is invested. The scheme survives exactly as long as new money keeps arriving, and it collapses the moment it stops, because there was never anything underneath it.

Every Ponzi scheme in history has collapsed. We tend to draw a moral lesson from this, which is that you cannot get something from nothing. That is not quite the lesson. The actual reason these schemes fail is more specific and more interesting. Each round needs more entrants than the last, and the supply of entrants is finite, and the participants keep leaving. People die, or retire, or need their money back. The scheme is a race between the growth of the obligation and the shrinkage of the pool, and the pool always wins.

Now take that same arrangement and write it into an economy that runs forever, populated by people who never exit.

The mathematics changes. Economists have known this since the 1950s, and it is not a fringe result. In certain models with infinite horizons, a bubble is not irrational and not doomed. It can be sustained indefinitely. Everybody in it behaves sensibly, everybody's expectations are met, and the thing never has to end. The formal name for these is rational bubbles, and the literature on them is respectable and old.

This is awkward, because economists would like their models to produce sensible prices, and a model that permits an asset to be worth infinity is not producing a sensible price. So the profession does something about it.

It assumes the problem away.

The assumption nobody looks at

The technical fix has a name. It is called a transversality condition, and if you have taken a graduate economics course you met it in the second or third week, in the middle of a derivation, presented as a piece of housekeeping.

What it says, stripped of the notation, is that the value of what you are still holding at the end must go to zero. You are not allowed to carry value off into the infinite distance. Everything has to be settled up eventually.

There is a companion assumption called the no-Ponzi condition, which says roughly that you cannot roll your debts forward forever, paying each one with a new one. Together they close the model. The infinities vanish. The prices behave. The bubbles are ruled out by fiat, and the derivation proceeds.

I want to be careful here, because this is not a scandal and I am not alleging one. These conditions are defensible. They usually correspond to something real, and generations of careful people have thought hard about when to impose them.

But notice what the assumption is actually doing. It is a statement that the party ends. That obligations have to be settled, that value cannot be deferred forever, that nobody gets to hold a claim into eternity without ever making good on it.

It is death, smuggled back into a model that had been allowed to forget about it.

And here is the claim this book is built on. That misbehavior in the infinite horizon models is not a modeling artifact to be patched. It is a preview. Those equations are describing, quite accurately, what happens in an economy of agents who do not exit, and we have spent seventy years treating the description as a nuisance rather than as information.

The model where death does the work

If that seems like a stretch, consider the model that runs the other way.

In 1958 Paul Samuelson published a paper describing what became known as the overlapping generations model. It is one of the load bearing structures of modern macroeconomics, and it is taught everywhere.

The setup is simple. People live for two periods. Young, then old. The young work and earn. The old do not work and need to consume. There is no way to store goods from one period to the next, so the old cannot simply save real output and eat it later.

So how do the old eat?

The answer is that the young give them things, and the young do this because they expect the next generation of young to do the same for them. That expectation is what money is, in this model. Money has value not because it is backed by anything, but because everyone believes the sequence will continue. It is a chain of obligations running forward through time, sustained entirely by the arrival of new participants.

If that sounds like the thing I described at the top of this chapter, it is because it is. The model shows that an arrangement we recognize as fraudulent when a private citizen runs it can be the foundation of a functioning monetary system when a society runs it, and the difference is that a society keeps producing new members.

Take a moment with what this means. One of the standard workhorses of the field is a machine whose engine is generational replacement. Its output depends on people being born, aging, and dying, in sequence, forever. Remove the turnover and the model does not merely become less accurate. It stops running.

So we have two families of models. One breaks when people stop dying. The other is powered by people dying. And in neither case does anyone treat mortality as a variable worth studying. In the first it is imposed as a technical assumption. In the second it is built into the plumbing. In both it sits there, unexamined, holding the structure up.

You cannot notice a variable that has never varied

Why has nobody looked at this?

Because until now there was nothing to look at. Mortality has been the most stable input in all of economics. Every other quantity the field studies has moved, often violently. Population, technology, energy prices, institutions, trade, the money supply, the size of the state. All of it swings around, and every swing generates a literature.

The human lifespan has not swung.

This deserves a moment of precision, because it is routinely misunderstood in both directions, and the misunderstanding matters for the rest of the book. Life expectancy at birth in wealthy countries has roughly doubled since 1850, from something in the thirties to something in the eighties. Some of that came from stopping children dying, and some of it did not: mortality fell at every age, and an adult today really does live substantially longer than an adult in 1850. Chapter 2 gives the numbers.

What has not moved is the ceiling. The longest verified human life still belongs to a Frenchwoman who died in 1997, and in the three decades since, with everything medicine has learned in the interval, nobody has come close.

So the number that determines the discount rate, the pace of inheritance, the turnover of ideas, the price of risk, and the rhythm of promotion has been, for the entire history of the discipline, a constant. Not a slow moving variable. A constant.

You cannot notice a variable that has never varied. You cannot run a regression on it. You cannot find its coefficient, because it has no variance to explain. It falls out of the analysis, not through carelessness, but because there is nothing there for the analysis to grip.

Which means that if the number ever does start to move, we will be looking at consequences with no theory of the cause. We will attribute them to policy, to culture, to central banks, to generational character, to anything at all except the thing that actually changed. In fact I think we are already doing this, and Chapter 3 is about the specific case where I believe it is happening in plain sight.

Five jobs

The argument of this book is that death is not simply an unfortunate fact about the human condition that economics has failed to model. It is functional. It is doing work. Remove it and specific machinery stops.

I count five jobs. The middle section of the book takes one chapter each.

It sets the price of time. The rate at which we discount the future contains a term for the chance you will not be there to collect. Lower the chance and the rate falls automatically, without anyone deciding anything. As it approaches zero, the value of anything permanent stops being a finite number.

It turns over capital. Inheritance is the largest redistributive event in any market economy, and nobody legislated it. Death is the only force that reliably breaks up a concentrated fortune, and it does so on a schedule, without exception, regardless of how well the fortune is defended.

It turns over ideas. Fields change direction when the people who defined them stop occupying the chairs. This is not a slur on scientists. It is a measured effect, and Chapter 5 goes through the evidence.

It prices risk. How much danger you will accept depends on how much life you are wagering. That is the subject of Chapter 6, and it produces the strangest result in the book, which is that a civilization becomes unable to leave Earth through exactly the process that makes leaving possible.

It creates openings. Careers are queues. Queues advance because people leave them. Every promotion in every hierarchy is somebody else's exit, and when the exits stop, so does everything downstream of them.

None of these five has a backup. Not one. We have no institution standing by to redistribute capital if inheritance stops, no mechanism to rotate intellectual authority if the holders never vacate, no procedure for opening a position that nobody has left. These functions were never designed, so they were never given redundancy. They emerged from a biological fact that was so reliable that nobody thought to ask what would happen if it changed.

What this book is not

Two disclaimers, both of which I will make good on in the next chapter.

This is not a book about whether we will live longer, and it does not depend on any particular biotechnology working. The argument runs on a much weaker premise than immortality, and I will state the premise precisely and defend it conservatively.

Nor is it a book about whether living longer is good. I think it is obviously good, in the way that not dying is obviously good, and I have no interest in the genre of essay that finds something spiritually improving about mortality. Death is not a teacher. It is a catastrophe that happens to be doing five jobs, and the jobs are the subject here.

What I am arguing is narrower and, I think, harder to dismiss. We are removing a beam. The beam was holding something up. Nobody has checked what.

The rest of this book is the inspection.

Chapter 2

Assume a Longer Life

This is the chapter where books like this one usually lose their serious readers.

The pattern is predictable. An author has an interesting argument about the consequences of some technology, and in order to get to the interesting part, they have to first establish that the technology is coming. So they write a chapter full of exponential curves and confident dates, and anyone with domain knowledge closes the book, because they can see the author has no idea how hard the actual problem is.

I would like to avoid that, and there is a straightforward way to do it. I am going to make my premise as weak as I possibly can, and then show that the argument still runs.

So let me start by listing what this book does not need.

It does not need immortality. Nothing here requires anyone to live forever, and Chapter 6 argues at length that forever is not on the menu regardless of what medicine achieves.

It does not need uploading, digital minds, or any transfer of a person into a machine. If that becomes possible it changes the argument substantially, and I flag where, but the argument does not rest on it.

It does not need a date. I am not going to tell you when, because I do not know, and neither does anybody else who is being honest with you.

It does not need any specific intervention to work. Not senolytics, not reprogramming, not rapamycin, not anything currently in a clinical trial.

What it needs is this. At some point, healthy human lifespans reach somewhere between 150 and 300 years.

That is the whole premise. Everything in this book follows from it, and it is a far weaker claim than the ones usually made in this genre. It requires roughly a doubling to a tripling of the current healthy span, which is less than the improvement in life expectancy at birth that the developed world already achieved between 1850 and 1950.

Now let me be honest about why that comparison, which sounds so reassuring, is actually the wrong one.

The two centuries that did not extend life

Life expectancy at birth in wealthy countries went from somewhere in the thirties in 1850 to somewhere in the eighties today. That looks like a doubling of the human lifespan, and it is routinely described that way.

There is a popular correction to this, which is that the whole thing is an artifact of child mortality. For most of history a large share of people died before their fifth birthday, and when you average a great many deaths at age two with a normal number at age seventy, you get a figure in the thirties that tells you very little about how long an adult actually lived. A Roman who reached twenty had a decent chance of seeing sixty.

That correction is popular, it is repeated constantly, and it is also wrong, or at least badly overstated. I believed it myself until I went and looked.

Mortality fell at every age, not just in childhood. Take the cleanest measure, which is to start the clock after the dangerous early years have already passed. In England in 1841, a five-year-old could expect to live about fifty-five more years, reaching sixty. Today a five-year-old can expect to reach eighty-two. That is a gain of more than twenty-five years, measured from an age where infant mortality has already been excluded from the arithmetic entirely.

So we did extend adult life, substantially, and anyone who tells you the whole story is dead babies is skipping the evidence.

Here is what actually did not move.

The longest verified human life belongs to Jeanne Calment, who was born in 1875 and died in 1997 at the age of 122 years and 164 days. She remains the only person ever verified to have reached 120, and she is more than three years clear of the next name on the list. Nearly thirty years have passed since her death, medicine has transformed, and nobody has come close.

That is the shape of the thing. We have gotten steadily better at delivering people to old age intact, and modestly better at extending old age once they arrive. What we have not done is move the far edge. James Fries named the resulting pattern in 1980 and called it the compression of morbidity: instead of dying steadily across the whole span of life, more and more of us stay healthy until we hit a wall in our eighties or nineties, and then decline quickly. Demographers describe the survival curve as becoming rectangular.

A rectangle has an edge, and that edge has not moved in the lifetime of anyone reading this.

This is why the reassuring historical comparison fails. The last two centuries of gains came from pushing more people up against the wall. The next doubling requires moving the wall, and moving the wall is not a harder version of the same task. It is a different task, and it has to be done somewhere we have never worked, which is inside the aging process itself.

What is actually on the table

So where does that stand? I will give the honest version, which is less exciting than the promotional version and more interesting than the dismissive one.

The serious framing arrived in 2013, when Carlos López-Otín and colleagues published a paper identifying what they called the hallmarks of aging. They listed nine. A revision in 2023 expanded it to twelve. The specifics matter less than the move, which was to stop treating aging as one undifferentiated process and start treating it as a list: genomic instability, shortening telomeres, changes in how genes are switched on and off, mitochondrial dysfunction, cells that stop dividing but refuse to die, and so on.

This converted a philosophical problem into an engineering one, because a list is something you can work through. And the work has produced real results. I am going to give them with their actual numbers, because the numbers are considerably more modest than the coverage, and the gap between the two is the most useful thing in this chapter.

Senescent cells. These are cells that have stopped dividing but refuse to die, sitting in tissue emitting inflammatory signals. In 2016 the van Deursen group showed that genetically deleting them in mice reduced age-related tissue dysfunction and extended lifespan. Drugs that do the same thing, called senolytics, followed. The most studied combination is dasatinib with quercetin, and in 2018 a study in Nature Medicine reported that it improved physical function and increased lifespan in old mice. Human trials exist and are early, small and aimed at specific diseases rather than at aging.

Partial reprogramming. This is the strangest result in the field. Shinya Yamanaka identified four genes that can revert an adult cell all the way back to a stem cell. In 2016 Ocampo and colleagues applied them in short pulses rather than continuously, two days on and five days off, which appeared to reset markers of cellular age without erasing the cell's identity, and without the tumors that continuous expression causes.

I want to be careful about what that study actually showed, because it is widely reported as more than it was. The large lifespan effect, roughly thirty percent, was in mice with progeria, a genetic disease of accelerated aging. That is a disease model, not normal aging. In ordinary old mice the demonstrated result was improved regeneration of damaged muscle and pancreatic tissue. Improved tissue repair in an aged animal is a genuinely exciting finding. It is not the same thing as making the animal live longer, and the study did not claim it was.

Rapamycin. In 2009 the National Institute on Aging's Interventions Testing Program reported that rapamycin extended lifespan in mice even when started at twenty months of age, which in mouse terms is late middle age. It was the first drug ever shown to extend lifespan in a mammal. The magnitude, from that late start, was about nine percent in males and thirteen percent in females.

Nine percent. Hold that number against the tone of most writing on this subject.

Metformin. A cheap, old, off-patent diabetes drug with enough suggestive observational data behind it that researchers proposed a formal trial, called TAME, to test whether it delays aging itself rather than any single disease. The design is six years, roughly three thousand adults aged sixty-five to seventy-nine, at a cost of around seventy-five million dollars.

It has not started. It has been awaiting funding for over a decade, and the reason is worth more attention than it usually gets. Because metformin is off patent, no company can own the result, so no company will pay for it. The most direct test we have proposed of whether a drug slows human aging is stalled not on science but on the absence of anyone with a commercial reason to run it. Anybody trying to forecast this field should hold that fact next to the enthusiasm, because it tells you what the incentives actually reward, which is proprietary molecules rather than answers.

Now the caveat that applies to all of it.

Almost every result above is in mice, worms or flies. The history of this field is full of interventions that worked beautifully in short lived animals and did nothing in people. Caloric restriction extends mouse lifespan by forty percent or more; in primates the effect is far smaller and in humans there is no comparable result. Researchers in the area have a rueful saying about having cured cancer many times over in mice.

There is a structural reason to expect this, and it is not merely bad luck. Short lived species are under less evolutionary pressure to maintain themselves, which leaves more cheap repair available to an intervention. We are already an unusually long lived mammal. Much of the easy maintenance has already been built into us, which means the remaining gains are the expensive ones.

So here is the honest summary. No intervention has ever been shown to extend the maximum human lifespan. Not one. There are real mechanisms, real drugs, real effects in animals, and a field that has become a science rather than a hope. There is also no human result, and anyone who tells you otherwise is selling something.

Why I am not going to give you a date

Given all that, the temptation is to pick a year, and I am not going to, for two reasons.

The first is that I would be wrong. Forecasts in this area have a dismal record in both directions. People have been predicting imminent breakthroughs in aging since the 1970s, and they were wrong. People also confidently declared that the maximum lifespan was fixed and that most cancers were untreatable, and they were wrong too. The honest position is wide uncertainty, and wide uncertainty does not compress into a date.

The second reason is more useful. The date does not matter for this argument.

Consider the two scenarios. Suppose the aging problem is solved in forty years. Then everything in this book is a description of the world your children negotiate, and the institutional questions in Part V are urgent.

Now suppose it takes four hundred years, or never happens at all. What have you actually lost by reading this?

You have lost nothing, because the five jobs are real right now. Death is setting the discount rate today. Inheritance is redistributing capital today. Scientific fields are reorganizing after their leading figures die today, and there is data on it. Careers are queues today, and gerontocracy is a live complaint in politics, in universities and on corporate boards today, with mortality still fully operative.

This book is, among other things, an argument that we already misunderstand a set of economic mechanisms because we cannot see the constant they rest on. That misunderstanding is causing errors now. Making the constant hypothetically variable is a device for seeing it, and the device works whether or not the hypothetical ever arrives.

So if you think the biology is a fantasy, read this as a thought experiment about the hidden structure of the present. The conclusions in Part II hold either way. Only the urgency changes.

The part that is already happening

There is one more reason not to wait for a breakthrough, and it is the reason I think this book is timely rather than speculative.

The effects do not require the full transition. They scale.

Every one of the five jobs depends on mortality continuously, not as an on and off switch. The discount rate contains a mortality term, so it moves whenever mortality moves, by however much. Inheritance flows depend on when people die, so they shift when the average shifts. The age at which authority vacates a chair is a number, and it drifts. Nothing here waits for a threshold to be crossed.

Which means that if my argument is right, we should already be able to see the early, mild version of it in the data. Not the dramatic version, but a measurable tilt in the predicted direction, over the decades in which healthy old age has been quietly extending.

We should see interest rates falling for reasons monetary policy does not fully explain.

We should see inheritance becoming a larger share of national wealth, and arriving later in the lives of the people who receive it.

We should see the average age of people holding senior positions rising, and the ladder underneath them getting slower.

We should see risk tolerance declining in wealthy societies, and safety regulation ratcheting in one direction only.

All four of those are documented trends. Every one of them has its own mainstream explanation, and I am not going to claim those explanations are wrong. I am going to claim they are incomplete, and that a single mechanism sits underneath all four, and that the mechanism is the one nobody looks at because it never used to move.

That is the case I will make over the next five chapters.

I want to have been clear about the strength of the claim, so let me restate it plainly one last time. I am not forecasting immortality. I am saying that mortality is an economic input, that it has begun to drift, that it may eventually move a great deal, and that we have no theory of what it does because we have never had to have one.

Assume a longer life. Not forever. Just longer. Then follow the money.

Chapter 3

The Price of Time

In 1648, a Dutch water authority responsible for maintaining a stretch of dike on the Lek river needed money for repairs. It did what borrowers did at the time. It issued a bond, written on goatskin, promising to pay interest to whoever held it.

Not for ten years. Not for a hundred. Forever.

That bond still pays. It was written out on the fifteenth of May, 1648, to a Mr Niclaes de Meijer, for a thousand Carolus guilders. One of the five known survivors is owned by Yale, which bought it for its finance archive in 2003 and periodically sends someone to the Netherlands to collect. In 2015 a curator came back with twelve years of arrears, amounting to about a hundred and thirty six euros.

The original rate was five percent. It was cut to three and a half, then to two and a half, in the seventeenth century, and two and a half is what it pays today. The institution that issued it has been reorganized more times than anyone has counted. The Dutch Republic that existed when the ink went onto the goatskin is gone. Three hundred and seventy odd years of European history have come and gone across that piece of skin, and the coupon has kept coming.

Britain did the same thing on a much larger scale. The consols, short for consolidated annuities, were government bonds with no maturity date. In February 2015 the Chancellor of the Exchequer redeemed 218 million pounds of four percent Consols, the first time in sixty-seven years that Britain had chosen to retire undated debt, and by that July every remaining undated gilt was gone from the portfolio.

Follow that particular debt backwards and it is worth the trip. The four percent Consols had been issued by Churchill in 1927, largely to refinance war bonds from the First World War. Those in turn had absorbed older obligations, including a bond Gladstone issued in 1853 to consolidate the capital stock of the South Sea Company, the enterprise whose collapse in 1720 gave us the phrase South Sea Bubble.

So a British taxpayer in 2015 made a final payment on a financial disaster that occurred under George the First. The debt had been rolled, renamed and refinanced across three centuries, and at no point in all that time did anyone have to redeem it, because it never came due.

I begin here because these objects make something visible that is otherwise hard to see. A perpetual bond has no maturity, which means the entire question of what it is worth reduces to a single number. Not the size of the payment. Not the creditworthiness of the issuer, at least not primarily.

The discount rate.

The most violent number in finance

Here is the arithmetic, and it is worth doing slowly, because almost everything in this chapter is a consequence of it.

Take an asset that pays you a thousand dollars a year, forever. What is it worth today?

The answer is the annual payment divided by the discount rate. That is the whole formula.

At a discount rate of ten percent, the asset is worth ten thousand dollars.

At five percent, twenty thousand.

At two percent, fifty thousand.

At one percent, a hundred thousand.

At a tenth of a percent, a million dollars.

At zero, there is no answer. Not a very large answer. No answer. The formula divides by zero and the value is not a number.

Look at the shape of that sequence. Moving from ten percent to five percent doubles the value. Moving from one percent to a tenth of a percent multiplies it by ten. The closer the rate gets to zero, the more violently the price responds to each further step down. The last few basis points matter more than everything that came before them.

Anyone who has traded long duration instruments knows this in their stomach rather than as a formula. It is why a thirty year bond moves several times as much as a two year bond on the same piece of news, and it is why pension funds with very long liabilities become almost hysterically sensitive to small changes in rates. Length amplifies. And a perpetuity is the limiting case of length.

So if you want to know what happens to the price of every permanent thing in an economy, you do not need to know very much. You need to know where the discount rate is going.

Where the discount rate comes from

Which raises the question of what the discount rate actually is, and why it sits where it does.

The standard framework goes back to Frank Ramsey, a Cambridge mathematician who wrote the founding paper on this in 1928 and then died two years later at the age of twenty six, which is either a bitter irony or an appropriate one for the subject of this book.

Ramsey's rule says the rate has two components.

The first is growth. If the future is going to be richer than the present, then a dollar delivered to that richer future is worth less than a dollar now, because the recipient will already have more. This part is about circumstances.

The second is pure time preference. This is the part that says a given amount of happiness matters less simply because it happens later. Not because the recipient will be richer. Just because it is further away.

Economists have always been uncomfortable with the second term, because stated plainly it sounds like a confession of irrationality. Why exactly should something count for less merely by being in the future? Ramsey himself called the practice ethically indefensible and said we only do it out of weakness of imagination.

But there is a reading of pure time preference that is not irrational at all, and it is the one this book cares about.

You might not be there.

A promise of a thousand dollars in fifty years is not the same object as a thousand dollars today, even if you are certain the money will be paid and certain of what it will buy, because there is a real chance you will not be alive to receive it. That is not a psychological failing. It is an accurate assessment of a probability.

This is not a rhetorical point I am imposing on the literature. It is in the models explicitly. Olivier Blanchard's widely used framework from 1985, sometimes called the perpetual youth model, handles mortality as a constant chance of dying in any given period, and that probability enters the effective discount rate directly. The higher your chance of death, the more heavily you discount. It is right there in the algebra, and it has been for forty years.

The most striking version of this appears in the climate debate, which I will come back to at the end of the chapter. When Nicholas Stern wrote his review of the economics of climate change for the British government, he had to choose a pure time preference rate, and his choice was famously low: one tenth of one percent. His justification was not psychological. He set the rate equal to the estimated probability that humanity ceases to exist.

He priced time as a mortality rate. That was not a metaphor. It was the actual construction.

So we have a chain, and every link in it is standard economics rather than anything I have invented. The value of every permanent asset depends on the discount rate. The discount rate contains a term for time preference. Time preference contains a term for the probability of not being there.

Extend life and you pull on the end of that chain.

The forty year decline nobody can fully explain

Now, the part of this chapter that is not theoretical.

Real interest rates across the developed world have been falling for roughly four decades. Not cyclically. Structurally. The measure economists use is r-star, the rate that would prevail with the economy at equilibrium, stripped of the business cycle and of whatever the central bank happens to be doing at the moment.

The standard estimates of it in the United States fluctuated between roughly two and two and a half percent from the 1990s through the mid 2000s, then fell to something near half a percent around 2009 and stayed there for years. Similar declines show up in Canada, the euro area and the United Kingdom, where recent estimates are the lowest in three decades.

This is one of the most consequential facts in modern finance and one of the least resolved. It has produced an enormous literature and a genuine puzzle, because the decline is too long and too broad to be explained by any central bank's policy. It preceded the 2008 crisis, survived it, and continued through wildly different policy regimes in different countries.

The explanations offered are several, and most of them have merit. A global savings glut.

Falling productivity growth. Rising inequality, since the rich save more. A shortage of safe assets. More recently, rates have risen again from their lows, which is a real development and which nobody should pretend away.

But one explanation appears in essentially every serious treatment, and it is demographics.

The mechanism is simple. People save for retirement. How much they need to save depends on how long they expect to be retired. As healthy old age extends, the required pile gets bigger, and the number of people in their peak saving years shifts. More desired saving chasing the same investment opportunities pushes the return on capital down. Work at the Bank of England attributed a substantial portion of the global decline since the 1980s to demographic factors alongside slower growth, and a body of academic work has since put numbers on the specific contribution of longer lifespans and shifting age structure.

I want to be careful about the strength of the claim, because this is the point in the book where a reader is entitled to be suspicious that I am reading my thesis into an ambiguous data series. So let me put it precisely. I am not claiming that longevity explains the fall in real rates. I am claiming something weaker and, I think, harder to argue with: that a mainstream, well documented body of research finds that longer lives push real rates down, that real rates have in fact gone down over exactly the period in which healthy old age has been extending, and that essentially nobody has asked what happens if the input keeps moving.

We have been treating this as a monetary policy story, because rates are what central banks do. Some of it is a monetary policy story. But underneath the cyclical noise there is a slow structural drift, and one of the things pushing on it is that people expect to be around longer.

Which means the transmission channel this book describes is not hypothetical. It is not something that switches on when a laboratory succeeds. It has been running, quietly, for forty years, and Japan is simply the country furthest along the curve.

What actually breaks

Push the rate further down and specific things stop working. Not metaphorically. Mechanically.

Pensions and insurance. A defined benefit pension is a promise to pay someone for as long as they live. The liability is long duration by construction, and a long duration liability is exactly the thing that explodes when rates fall. Every pension fund in the developed world already knows this, because it has spent the last two decades watching its deficit widen for reasons that had nothing to do with its investments. Now extend the payout period. The liability grows because the payments last longer, and the present value of each payment grows because the discount rate is lower, and both effects run the same way at the same time. Life insurance runs into the mirror image of the same problem, which is that a product priced on an actuarial table is a product priced on an assumption about when people die.

Anything permanent. Land, water rights, mineral rights, spectrum, orbital slots, the freehold on a building. These are perpetuities that happen to be made of dirt and paper. As the rate falls, their prices rise faster than the rate falls, and there is no ceiling in the formula. We have watched a version of this already: a substantial part of the rise in developed world house prices over recent decades is not a story about construction costs or about bricks.

It is the same asset being capitalized at a lower rate. The building depreciates. The land underneath it is the perpetuity, and the perpetuity is what moved.

The idea of a fair price. This is the deepest of the three and the least discussed. Valuation is a technology for compressing an unbounded future into a single present number, and it works because the far future contributes almost nothing. Discount a payment eighty years out at eight percent and it is worth about two tenths of a cent on the dollar. It vanishes, which is convenient, because it means you do not have to know anything about the year 2106 in order to price something today. That convenience is a gift from the discount rate. Take the rate toward zero and the gift is withdrawn. The year 2400 starts to matter as much as next year, and no one has any idea how to price the year 2400.

The argument we are already having

There is one place where this fight has already broken into public view, and almost nobody noticed what it was really about.

In 2006 the Stern Review concluded that the world should spend heavily and immediately on climate change. William Nordhaus, who later won a Nobel for his work in the same area, concluded that the world should spend considerably less and more gradually.

The two men were not really disagreeing about climate science. They were disagreeing about the discount rate. Stern used a very low one. Nordhaus used a higher one. Because the costs of climate change land mostly on people in the distant future, and the costs of preventing it land on people now, the entire policy recommendation swings on that single parameter. Change the rate and the answer flips, with the physics held constant.

The debate that followed was fierce, and it was conducted in the language of ethics. What do we owe future generations? Is it defensible to weight a person less because they were born later?

Now put this book's premise on top of that debate and watch what happens.

Stern's low discount rate was an ethical assumption, contested precisely because it was an assumption. He was asking us to behave as though we cared about the far future roughly as much as the near one, and his critics said that was not how people actually behave.

If lifespans extend, that assumption stops being ethical and becomes descriptive. You do not have to be persuaded to care about the year 2200 out of moral obligation to strangers. You will care about it because you plan to be there, and so will everyone you are negotiating with.

I find this the single most interesting consequence in the chapter, and it cuts in an unexpected direction. Long lives may be very good for climate policy and for anything else that requires patience, because they align private incentives with the long run in a way that no amount of ethical argument has managed. The same force that breaks the pension system might build the sea walls.

That is worth holding onto, because this book is going to describe a great deal of machinery seizing up, and the honest accounting includes the things that get better. A civilization that discounts the future at nearly zero will build things we cannot currently justify building. Cathedrals, in effect.

It will also find that it has no idea what anything is worth.

The price of time

Everything in economics that involves the word "worth" runs through the discount rate. It is the exchange rate between now and later, and like any exchange rate, it is a price rather than a law.

We have been treating it as a law. We have built pension systems, insurance markets, valuation methods, regulatory cost benefit analysis and the entire apparatus of capital allocation on top of a number that we inherited rather than chose, and one of the things determining that number is how long the people using it expect to live.

That number has already started to move. The next four chapters are about the other four things it was holding up.

Chapter 4

The Estate

In 1961 the Supreme Court of California decided a case called Lucas v. Hamm, and in doing so it said something no court likes to say about the law.

The facts were unglamorous. A lawyer named Hamm had drafted a will for a man named Emmick. The will contained a provision that violated an old rule of property law, the provision was declared invalid, and the intended beneficiaries ended up settling with the family for seventy five thousand dollars less than the will would have given them. They sued the lawyer for malpractice.

The court held that the lawyer was not liable. Not because he had drafted it correctly, but because the rule he had broken was so notoriously difficult that a reasonably competent attorney could not fairly be expected to get it right.

The rule in question is called the rule against perpetuities. Law students dread it. It is the standard example of a legal doctrine that defeats intelligent people, phrased in a formulation that has to be read several times before it stops being a sentence and starts being a meaning: no interest in property is valid unless it must vest, if it vests at all, no later than twenty one years after the death of some person alive when the interest was created.

Read that again, slowly, and notice what it is actually about.

It is about the dead letting go.

Four hundred years of prying fingers loose

The rule emerged from an English case in 1682 concerning the Duke of Norfolk's family, and the problem it was invented to solve is one that recurs whenever people become both wealthy and worried.

A rich man wants to control his property after his death. Not just to leave it to someone, but to dictate its use in perpetuity: this estate shall never be sold, it shall pass always to the eldest son, it shall be used for these purposes and no others, forever. The instruments for doing this existed. Trusts, entails, settlements. With enough legal ingenuity, a landowner could bind his property to his intentions for centuries.

English courts decided this was intolerable, and their reasoning was economic rather than sentimental. Property that cannot be sold cannot move to whoever values it most. Land locked in a dead man's instructions falls out of the market and stays out. A country in which enough estates are tied up in perpetuity is a country where the productive use of land is set by people who died before anyone currently alive was born.

So the courts drew a line. You may reach into the future, but only so far. Roughly, you may control your property for the lifetimes of people who are alive when you write the document, plus another twenty one years for their children to reach adulthood. After that, it belongs to the living.

That is what all the tortured phrasing is protecting. A time limit on the grip of the dead.

And here is the thing I want to fix in your mind before the rest of this chapter, because everything follows from it. The rule against perpetuities exists because death alone was not sufficient. Even with mortality operating at full strength, even with every owner reliably dying on schedule, wealthy people found ways to project control forward, and society had to construct an additional legal mechanism to stop them.

That is the baseline. Four hundred years of specialized law, needed to force turnover in a world where everyone dies.

Now remove the dying.

The largest redistribution nobody voted for

Inheritance is the biggest transfer of wealth in any market economy, and it happens without legislation, without political fights, and without anyone being able to opt out.

Think about what death does to a fortune. It stops the strategy. The person who built the position, who understood the assets, who had the nerve and the relationships and the specific view of the world that made it work, is no longer there. The estate passes to people who did not build it. It gets divided among heirs, which cuts each piece. It gets taxed. It gets managed by someone with different judgment, different appetite, and usually less obsession.

Every culture has a proverb for what happens next. In English it is shirtsleeves to shirtsleeves in three generations. In Lancashire it was clogs to clogs. The Chinese version says wealth does not survive three generations. These sayings exist everywhere because the phenomenon is everywhere, and the phenomenon has a mechanism, and the mechanism is death.

Thomas Piketty's central observation is that when the return on capital exceeds the growth rate of the economy, wealth concentrates. Capital compounds faster than everything else, and the gap between the owners and everyone else widens. The formulation is usually written as r greater than g, and it has become one of the most argued about inequalities in modern economics.

But notice that the concentration Piketty describes is slow. It takes generations to become socially visible, and it repeatedly reverses. His French data make the shape of it plain. The annual flow of inheritance ran at twenty to twenty-five percent of national income from 1820 through 1910. By 1950 it was under five percent. By 2010 it was back to about fifteen.

That is not a ratchet. It is a U, and something drove it down to the bottom of the U before it started climbing again.

What interrupts it?

Wars and depressions, in the twentieth century case, which destroyed capital on a scale nothing else has matched. But underneath those episodic catastrophes runs the constant one. Every fortune is handed, on a schedule set by biology, to someone who did not make it.

Seventeen thousand times

Let me make the compounding concrete, because the number is more startling than the argument.

Suppose you have capital earning five percent a year after inflation. That is not an aggressive assumption. It is roughly what a diversified portfolio of productive assets has delivered over long historical periods.

Over thirty years, a working career, that multiplies your money by about four and a third. Respectable. This is the arithmetic of a successful life.

Over two hundred years, the same five percent multiplies your money by about seventeen thousand.

Not seventeen thousand dollars. Seventeen thousand times. A million becomes seventeen billion, in real terms, at an unremarkable rate of return, with no brilliance required at any point. Just continuity.

That number is the entire chapter. Compounding at ordinary rates produces absurd outcomes over extraordinary periods, and the only reason we do not observe absurd outcomes is that nobody has ever had an extraordinary period. The compounding always gets cut. The owner dies, the estate splits, the tax lands, the heir sells the business to buy a house in the south of France.

Remove the interruption and r greater than g stops being a slow drift that societies argue about. It becomes a fixed point. Capital accumulates without limit in the hands of people whose strategies never change, whose risk appetite never resets, and who never hand anything to anyone.

And the resulting distribution is not merely unequal. It is frozen. That distinction matters more than it sounds. Societies tolerate a great deal of inequality when the positions turn over, because there is a story available in which the arrangement is provisional. Remove turnover and you have removed the story, and what is left is not a market outcome. It is an aristocracy, in the original and precise sense of the word: a permanent stratum, defined by birth, that cannot be entered.

The machinery we already dismantled

Here is the part that should make you uncomfortable, because it is not a projection. It already happened.

Beginning in the 1980s, a number of American states abolished or drastically extended their rule against perpetuities. South Dakota went first, in 1983. Delaware, Alaska, Nevada, Idaho, Wisconsin and others followed. Some eliminated the rule outright. Others replaced it with limits so long, several centuries, that they amount to the same thing.

The motive was not ideological. It was competitive. Trusts generate fees, and fees generate jobs and tax revenue, and a state that lets a trust run forever will attract trust business from states that do not. There was a tax dimension too. Federal changes in 1986 created a strong incentive to shelter wealth in structures that could skip generations, and a trust that never terminates is the ideal vehicle for that purpose.

So a race began, and it was a race to see which state would let the dead hold on longest.

The product this created is the dynasty trust: a structure that holds assets in perpetuity, distributing to descendants indefinitely, never passing through anybody's estate, and therefore never triggering the transfer taxes that estates trigger. Assets held under these arrangements in the leading states now run into the hundreds of billions of dollars, and a series of journalistic investigations over the past few years has made the scale of it public.

I want to state the sequence plainly, because it is the most consequential thing in this chapter.

For four hundred years, common law jurisdictions maintained a rule whose only purpose was to force property to return to the living. In the space of about thirty years, and for reasons that had nothing to do with anything in this book, a significant part of the developed world took that rule apart in exchange for trust management fees.

We disabled the anti permanence machinery first. The permanence is arriving second.

Firms die too, and that is the point

The same argument runs through the corporate sector, and it is easier to see there because the data is public.

Companies are mortal. They fail, they get acquired, they get broken up, they slowly become irrelevant and are quietly removed from the index. The average tenure of a company on the S&P 500 was about thirty-three years in the mid 1960s. By 2016 it was around twenty-four, and the forecasts have it heading toward the low teens within a few years. On current churn, roughly half the index turns over in a decade.

Economists call this creative destruction, after Joseph Schumpeter, and it is understood to be the mechanism by which capitalism reallocates resources. New firms with better methods take capital, customers and staff away from old firms with worse ones. The old firms die. This is not a failure of the system. It is the system.

Notice that the process requires the death of institutions, and that the death of institutions has historically been tied to the mortality of the people running them. Founders age. They lose their edge, or their nerve, or their grip on what customers now want. Succession is forced on them, and succession is where a great many powerful companies lose their way, and that losing of the way is what makes room for the next thing.

Two developments are now pushing against this at once.

The first is already here. Dual class share structures, in which founders hold shares carrying ten or more votes each, have become normal in technology listings over the past two decades. The explicit purpose is to insulate the founder's control from the market's opinion, permanently. The founder can be outvoted by no one.

The second is the subject of this book. Combine permanent control with a founder who does not age, and the concept of corporate succession simply ceases to have a referent. There is no handover, because there is no one to hand to and no occasion that forces it.

A world where firms do not die is a world where creative destruction is missing its second half. We would keep the creation and lose the destruction, which sounds like an improvement and is not, because the destruction is how the resources get released.

What death was doing here

Let me summarize the job, since this chapter has covered a lot of ground.

Death performs a compulsory, universal, unappealable redistribution of capital on a schedule that nobody controls. It cuts every accumulation before the compounding gets absurd. It transfers assets to people who did not build them and mostly cannot maintain them. It forces institutions into succession crises that open room for competitors. It does all of this without a vote, without a policy, and without any way for the wealthy to opt out, which is the feature that four centuries of ingenious lawyers have been trying and failing to engineer around.

Nothing else in our institutional repertoire does this. Estate taxes are a pale imitation, they are avoidable, and they are politically fragile precisely because they are visible as choices in a way that mortality is not.

We have no backup for this function. And unlike the discount rate, which will drift gradually and give us time to notice, this one has a sharp edge. The moment the first cohort of very long lived, very wealthy people exists, the redistribution stops for them and continues for everyone else.

That is not a story about the year 2200. Given how these things distribute, it is a story about a few thousand people, quite early, and the rest of us watching.

Chapter 5

The Funeral Principle

Max Planck, who founded quantum theory and spent his life inside the institutions of German science, wrote near the end of it that “a new scientific truth does not triumph by convincing its opponents.” They do not come round, he observed. They die, and a generation grows up for whom the new idea was never controversial.

The line is usually quoted at conferences, and it usually gets a laugh. Everybody in the room can think of somebody it applies to. Then the session moves on, and nobody treats it as what it is, which is a testable empirical claim about how knowledge actually propagates through a profession.

It has been tested. The result is not funny.

Four hundred and fifty two deaths

The study that did it comes from Pierre Azoulay, Christian Fons-Rosen and Joshua Graff Zivin, and the design is grim and elegant. They identified four hundred and fifty two elite life scientists who died prematurely while still active in their fields. Not retirements, not slow declines. Sudden exits, which is what makes the design work, because an unexpected death is close to a natural experiment. It removes a person from a field without any of the gradual withdrawal that would otherwise muddy the picture.

Then they looked at what happened to those fields afterward.

Two things did. The scientist’s close collaborators published meaningfully less, which is unsurprising and mostly reflects the loss of a productive partner. But the second finding is the one that matters. Researchers who had never worked with the deceased, outsiders to that particular circle, began publishing in the area at a higher rate. The increase averaged 8.6 percent, and it did not fade.

And their papers were different. They drew on different prior work. They were disproportionately likely to be highly cited. The authors were disproportionately people who had not previously been active in that subfield at all.

The paper’s own conclusion is blunter than anything I would write. Once in control of the commanding heights of a field, star scientists tend to hold their position somewhat too long.

The ideas had been out there the whole time. What changed was that the position blocking them became vacant.

I want to be careful about the accusation embedded in this, because the obvious reading is uncharitable and I do not think it is correct. The finding is not that eminent scientists are villains suppressing rivals. Some are, but that is not what the data shows and it is not necessary for the effect.

The mechanism is much more ordinary, which is what makes it so hard to fix.

How a field gets held

Consider what an eminent researcher actually controls, not by scheming, but by holding the job.

They sit on the panels that award grants. They referee for the journals that matter, and they edit some of them. They advise on hiring and tenure. They train the graduate students who become the next cohort, and those students learn the field's questions from them. They are asked to write the review articles that define what the open problems are. When a science journalist wants a quote, they get the call.

Now suppose this person sincerely believes, on the basis of forty years of hard work and genuine expertise, that a particular line of inquiry is a dead end.

They are not going to suppress it. They will simply rate that grant application slightly lower, because they honestly think it is less promising. They will recommend rejection of that paper, because they honestly see the flaw in it. They will advise a promising student against that topic, because they honestly want the student to have a career. Every one of those judgments is made in good faith. Every one of them is exactly what expertise is supposed to be for.

And the aggregate of those good faith judgments is a field that cannot turn.

This is why the effect is so robust and so difficult to legislate against. There is no bad actor to remove. There is only the ordinary operation of authority, which is that people with track records get to decide what counts, and the track record was built in the old paradigm.

The history is full of it, and three cases are worth having in mind because they fail in three different ways.

Ignaz Semmelweis worked out in the 1840s that doctors were killing mothers by walking from the autopsy room to the delivery room without washing their hands. In 1847 he ordered handwashing in chlorinated lime at the Vienna General Hospital and the death rate in his ward fell from around eighteen percent to about two.

That is not a subtle statistical finding. That is roughly eight out of every nine deaths, gone, in twelve months. The profession rejected it anyway, and part of the reason is that his result was not merely a hypothesis but an accusation. It said that the physicians hearing it had personally been killing their patients, for years, with their own hands. There is no version of that claim that a room of eminent men can weigh dispassionately. Semmelweis was committed to an asylum in 1865, was beaten by the guards there, and died of an infected wound.

Alfred Wegener presented his case that the continents move to the German Geological Society in January 1912. He was rejected for roughly half a century, and here the mechanism of rejection was different and considerably more defensible. He could not explain what force would push a continent through solid ocean floor, and his critics were right that he could not. He was also a meteorologist rather than a geologist, which meant the people qualified to judge his claim were the people whose field he had wandered into. Continental drift became consensus in the 1960s, once seafloor spreading supplied the missing mechanism. Wegener died on a Greenland expedition in 1930, three decades before he was vindicated.

Both things were true at once in his case. The objection was legitimate and the objectors were also incumbents, and nobody can now cleanly separate how much of that fifty year delay was proper scientific caution and how much was territory.

That is the uncomfortable part. In the Semmelweis case we can say cleanly that the field was wrong. In the Wegener case we cannot, and most real cases look like Wegener rather than Semmelweis.

Barry Marshall is the third kind. He could not persuade anyone that stomach ulcers were caused by bacteria rather than by stress and acid, so in 1984 he drank a culture of the organism and gave himself gastritis. He received the Nobel Prize in 2005.

Marshall found a way around the problem that did not require waiting for anybody to die. It required an act so dramatic that it could not be refereed away, could not be declined by a grant panel, and could not be politely ignored at a conference. That worked. It is also not a mechanism. You cannot build a scientific establishment on the assumption that people with correct unpopular ideas will poison themselves to get a hearing, and the ones who are unwilling to are not thereby wrong.

The honest counterweight

I should state the cost of turnover directly, because a chapter that only lists the benefits of people dying is not an honest chapter.

When a great scientist dies, an enormous amount is lost that was never written down. Judgment about which problems are worth attacking. Knowledge of which apparently promising avenues were tried in 1987 and quietly failed. The tacit craft of doing the work, which passes from person to person and does not survive in papers. Fields do not only get stuck by keeping their elders. They also get stupid by losing them, and rediscover things at great expense that somebody already knew.

So the claim here is not that death is good for science. It is not. The claim is narrower and it is the same claim as every chapter in this section.

Death is currently performing a function, badly and at enormous cost, and we have no other mechanism that performs it at all.

Notice also that our existing institutions had a partial substitute and then removed it. Mandatory retirement for tenured faculty in the United States ended on the first of January, 1994, and academic careers stopped having a defined endpoint. I should say that the consequences are genuinely disputed. The higher education establishment predicted that senior faculty would stay indefinitely and strangle junior hiring, and at least one careful study concluded beforehand that the effects would be far smaller than feared. Retirement rates at seventy and seventy one did fall sharply afterward, which is not in question. Whether that translated into the broader damage that was predicted is still argued about.

I raise it not as proof but as the only real experiment we have. One profession removed one turnover mechanism, deliberately, and we are still arguing about what it did. That is a poor evidentiary position from which to remove all five of them.

The compounding problem

Now put this chapter next to the previous one, because the interaction is worse than either alone.

Chapter 4 argued that capital would stop turning over. This chapter argues that intellectual authority would stop turning over. In practice these are not two separate populations. The people who hold capital and the people who hold intellectual authority overlap heavily and increasingly. Endowments fund research. Philanthropic foundations set agendas. Private laboratories dominate certain fields outright, and the person who founded the lab decides what it works on.

Freeze both at once and you do not get two static systems side by side. You get one static system with mutual reinforcement. The capital funds the ideas that the authorities endorse. The authorities are the people whose prestige was built with that capital. Neither has any mechanism for revision, because the revision mechanism in both cases was the same event.

There is a third layer arriving, and it deserves flagging here even though it belongs properly to Part III.

Machine learning systems are trained on the accumulated written output of a field. That corpus is produced disproportionately by the people with the most standing, because standing is what gets you published, cited and quoted. If those people never leave, they never stop producing, and their share of the corpus never declines. The models trained on it will reproduce the consensus of a generation that is not being replaced, and they will do so with fluency and at volume, and their output will in turn become part of the corpus.

Every previous generation's intellectual authority had a hard stop. Yours ended when you did, and even the greatest reputations decayed as the people who remembered you personally also went. What we are building is the first apparatus for holding a position in a debate indefinitely, at scale, without the holder needing to be present.

Combine that with a holder who is present anyway.

What death was doing here

The job is straightforward to state and impossible to replace with anything we currently have.

Death vacates the chairs. It empties the grant panels, the editorships, the department heads, the named professorships and the foundation boards, on a schedule that nobody controls and no amount of eminence can defer. It does this without any judgment about whether the occupant deserved to go, which is precisely the property that makes it work, because any mechanism that required a judgment would be captured by the people being judged.

And it does one more thing that is easy to miss. It supplies the young with a credible expectation that the positions will eventually be available. That expectation is what makes it rational for a talented person to enter a field at all, to spend a decade in training for a chance at a chair. Remove the expectation and you have not merely blocked the ladder. You have removed the reason to start climbing.

Which is the subject of Chapter 7. But there is a chapter in between, because before we get to what happens when nobody leaves their post, there is a stranger question about what happens when nobody is willing to take a risk.

Chapter 6

Nobody Goes to Space

In July 1969, a White House speechwriter named William Safire wrote a speech that was never delivered. It was for President Nixon, and it opened like this: "Fate has ordained that the men who went to the moon to explore in peace will stay on the moon to rest in peace."

The plan around it was worked out in detail. Nixon would telephone the widows first. Then a clergyman would commend the men's souls to the deep, the way it is done at a burial at sea. Then Mission Control would close the line. Two men would suffocate on the surface of another world while the entire planet listened.

The speech went into a drawer. NASA lit the rocket anyway.

We have stopped finding this remarkable, and it is the only part of the Apollo story that matters for what comes next. Armstrong himself said later that he had privately put his chances of getting back to Earth at about ninety percent, and his chances of landing successfully on the first attempt at no better than even. Nobody on that program was confused about the arithmetic. They ran the numbers, they wrote the eulogy, they filed it, and they launched.

The usual explanation for this is character. People were braver then. The culture was harder. Test pilots were a particular breed of person.

I want to offer an explanation that is less flattering and a good deal more useful.

They were cheaper.

Not cheaper as people. Cheaper as assets. The economic value of a human life is not a constant of nature. It is a price. Prices move, and over the next century every force we are excited about is going to push this one in the same direction: up, steeply, and with nothing to stop it.

The frontier will not close because we lack the engines. It will close because we can no longer afford the funeral.

What a life costs

Regulators have to put a number on a life. People find this offensive when they first hear it. Then they think about it for a minute and realize there is no way around it.

Suppose a highway guardrail costs forty million dollars and prevents one death every other year. Somebody has to decide whether to build it. Refusing to name a number does not spare anyone that decision. It just makes the decision worse, and hides who made it.

So the number gets named. It is called the value of a statistical life, and the US Department of Transportation put it at 13.7 million dollars for 2024, up from 9.1 million in 2012. The Department of Health and Human Services uses a figure close to 13 million. The agencies differ, and the number climbs every year.

It is not the value of you. Nobody is claiming that. It is built from what large numbers of ordinary people actually demand, in wages or in cash, before they will accept a small increase in their chance of dying. It comes out of what people do, not out of anyone's philosophy. Dangerous jobs pay more. Economists measure how much more, and work backward.

Two things about that number deserve more attention than they get. They are usually treated as footnotes. They are actually the whole story.

It goes up with wealth. Richer people demand more money to accept the same danger. This is not a claim about whose life counts. It is simple arithmetic about what a dollar is worth to you. When you have very little, an extra thousand dollars changes your life, and you will accept real risk to get it. When you have a great deal, an extra thousand dollars changes nothing, and the same risk is no longer worth considering. The value of a statistical life is really a measure of how much the world still has left to offer you.

It goes up with time remaining. The standard move here is to convert the lump sum into a value per year of life, by dividing it across the years a person can expect to have left. Do that and you land somewhere in the low hundreds of thousands of dollars per year. Roughly that figure sits behind how most rich countries decide which medical treatments to pay for.

Now run it backward. If a year of life has a price, then a life is worth that price multiplied by the number of years in it.

There is nothing in that equation that stops.

Two bonds and a perpetuity

Here is the same idea in the language of the bond market, where I think it becomes obvious.

Think of a human life as a stream of future years, the way a bond is a stream of future payments. A seventy-year-old is a short bond. A few more payments, then it matures. A healthy twenty-five-year-old today is something like a fifty-year bond. Both are finite. Both eventually stop on their own.

Now cure aging. Not death. Aging.

The person in front of you is no longer a fifty-year bond. They are a perpetual bond, one that never matures and never stops paying.

Anyone who has priced a perpetual bond knows what happens next, and knows it in their hands rather than as a theorem. The thing becomes wildly sensitive to small changes in the inputs. Traders call this duration and convexity. The plain version is easier: when the payments never stop by themselves, the only question that matters is whether something might stop them by accident.

That is the entire valuation. All of it. Every bit of what a perpetuity is worth lives in the chance that it gets interrupted.

Which is why the character explanation of Apollo misses the point. Armstrong was not braver than a person who will live a thousand years. He was shorter dated. He was betting thirty or forty expected years on a coin flip. The thousand-year person is asked to bet a perpetuity on the same coin flip. No amount of courage changes what that trade is worth.

What kills you when nothing else does

It is worth being exact about how long "indefinitely" actually is, because the answer is not forever, and the gap matters enormously.

Human mortality follows a curve that Benjamin Gompertz described in 1825. After about age thirty, your annual chance of dying doubles roughly every eight to ten years, depending on the population you measure. That is the aging term. It is why a healthy sixty-year-old and a healthy thirty-year-old are, statistically, completely different animals. Nearly everything we call medicine is an argument with this one curve.

Sitting underneath it is a second, much flatter risk that has nothing to do with aging at all. Cars. Falls. Fires. Drownings. Violence. Aircraft. This is the rate at which the physical world removes people who were in perfect health that morning. In a wealthy country, for a young adult, it runs somewhere on the order of one in two thousand per year.

Now suppose we win. Suppose the aging term goes to zero and only the background risk is left. Your life expectancy becomes the simple flip of that number. One in two thousand per year gives you something on the order of two thousand years.

I should be clear that this is my own arithmetic rather than a finding I am reporting. It is not a demographic projection and no one has established it. It is one line of division, and I am doing it in front of you precisely so you can see how thin it is. Make cars and medicine safer and the answer climbs. Let the world get more dangerous and it falls fast. The whole estimate rests on a single input that almost nobody thinks about, which is itself the point of the chapter.

But look at what happened to the shape of risk, because this is the part that should stop you.

Today, aging kills you and accidents are a rounding error. In the world we are describing, aging kills nobody and accidents kill everyone. One hundred percent of death becomes accidental.

Every single death is now, in the strict sense, a preventable one. Somebody chose to run that risk, or allowed someone else to run it. No society in history has had to live under that fact. Every workplace fatality, every car crash, every launch failure stops being a tragedy and becomes an accusation. Somebody signed off on that hazard.

You cannot die and get alive again

Now the formal core of the chapter. I will put it as plainly as I can, because everything else rests on it.

You can go broke and get rich again. You cannot die and get alive again.

That asymmetry has a name in probability. Death is an absorbing state. Once the process lands there, no future period exists in which to recover. Every other setback is a detour. This one is a wall.

Anyone who has thought seriously about position sizing knows where this leads. It is the whole content of the Kelly criterion, and of what Ole Peters has more recently formalized under the name ergodicity economics. The idea is this. Imagine a bet with good expected value that also carries a one percent chance of wiping you out. Give that bet to a thousand people once each, and the group does beautifully. Give it to one person a thousand times in a row, and that person is broke with near certainty.

Same bet. Completely different outcome. Averages across a crowd tell you nothing about what happens to a single person playing in sequence, and the difference is entirely caused by the fact that ruin has no exit.

Now apply that to a life with no natural end. Take any fixed yearly chance of dying by accident, however small. Survival over many years means multiplying that small survival chance by itself, over and over. Any number below one, raised to a high enough power, goes to zero.

Over a long enough horizon, ruin is not a risk. It is a certainty with a waiting time.

Which produces a conclusion I find genuinely startling, and which I have not seen stated anywhere in the longevity literature:

Curing aging does not give you immortality. It converts immortality from a biology problem into a risk management problem, and the risk management version is harder.

Biology is a finite opponent. There is a specific list of mechanisms, and in principle each one can be addressed. Accidents are not a finite opponent. They are the open-ended set of ways a physical universe can intersect a fragile body, and driving that to zero is not a research program. It is a religion. The best anyone can do is push the number down, decade after decade, forever, and every further reduction costs more than the last, because you are working through a longer and longer tail of stranger and stranger ways to die.

So the rational policy for a person who does not age is not to live well. It is to reduce the hazard, continuously, at almost any price. Not out of cowardice. The arithmetic simply does not offer another answer. Every unit of risk you accept is no longer a bet against forty remaining years. It is a bet against all of them.

The price of the frontier

Now put the pieces together.

Space is, and will be for a long time, a place with an unavoidably elevated chance of dying. Not because engineering cannot improve, since it improves enormously, but because the floor is set by physics rather than by care. Rockets are controlled explosions. Vacuum does not forgive a single broken part. Radiation accumulates, shielding is heavy, and mass is the constraint on everything. Journeys anywhere interesting take a long time, and if something goes wrong halfway, the options range from poor to none.

Consider the record. The Space Shuttle lost two vehicles and their crews across a hundred and thirty five flights. Call it one and a half percent per flight. NASA's own final risk assessment put the mean figure at about one in ninety, which is consistent with what actually happened. That was the most sophisticated program the richest country on earth could mount, over thirty years, with the lessons of the first disaster already absorbed.

Now be generous to the future. Suppose the next century does spectacularly well and gets that down to one in ten thousand. Per journey, that is safer than plenty of things people do without a second thought.

For someone with forty years left, one in ten thousand is nothing against the chance to be among the first humans somewhere. It is no worse than a career flying small aircraft, and people take that deal happily.

For someone with two thousand years left, that same one in ten thousand is a bet of about a fifth of a year of expected life. Which sounds small, until you price it. At a few hundred thousand dollars a year, you are already looking at tens of thousands of dollars of risk premium per person per launch, before anything else is counted. And remember that the price of a life-year is not fixed either. It rises with wealth, and this is by construction the wealthiest society that has ever existed. More years, and each year worth more. Multiply those together and the cost does not rise in a straight line. It compounds.

That is only the direct cost. The real closure happens somewhere else, and it happens without anybody deciding it.

Insurers price the tail, and the tail is now enormous. Liability follows the insurance. Regulators follow the liability, and they are already cautious, and they now answer to a public for whom every death is somebody's fault. Then there is capital, and here I am describing my own trade. The people with balance sheets large enough to fund frontier ventures are precisely the people with the longest personal horizons and the most to lose. They are also, for the first time in history, going to be personally present when the consequences arrive. A hundred-year project is no longer something you hand to your successors. It is something you will still be standing next to when it fails.

Nobody bans space travel. The insurance simply becomes unwritable, and the whole thing quietly dies of underwriting.

So here is the paradox in its final form.

A civilization becomes able to settle the frontier through exactly the same process that makes it unwilling to. The wealth, the medicine and the safety that make the stars thinkable are the wealth, the medicine and the safety that make the cost of reaching them unbearable. Ability and willingness are not two forces in tension. They are one force, running in opposite directions.

Every civilization rich enough to reach the stars is too rich to go.

Four objections

Let me take the strongest versions of the counterarguments, since a paradox that survives only the weak ones is just a slogan.

It only takes a few volunteers. True, and beside the point. Frontier settlement has never been short of willing bodies. What it runs short of is capital, insurance, launch licenses, supply chains and legal cover, all of which sit with people who want nothing to do with the tail. Frontiers are opened by financing, not by volunteers. Magellan's expedition left in 1519 with five ships and around two hundred and seventy men. One ship and eighteen men came home three years later. The reason it sailed at all is that the Spanish crown could absorb a loss it fully expected to take. The volunteers were never the constraint. The underwriter was.

Backups and copies dissolve the problem. This is the serious objection, and I think it is correct, for anything that can actually be copied. If what you are can be duplicated and stored, then death stops being a wall and becomes an expensive inconvenience, and everything I have argued collapses. Notice exactly what that concedes. The frontier reopens only for things that can be copied. Which is not us. That takes me directly into the next chapter.

Machines will go instead. Yes. That is my argument, not a refutation of it. Robotic and artificial explorers are the obvious answer to an intolerable death rate, and they will do the work. But we should be clear about what follows. Whoever bears the risk is whoever establishes a presence, and presence, historically, is what turns into ownership. The frontier will be settled by whatever can afford to die out there. It will be owned by whatever settles it. Whether those remain the same thing is the political question of the next two centuries.

The old and the sick will still go. They will. And this is the objection that turns the argument rather than defeating it.

The mortal inherit the stars

If willingness to take a risk depends on how much life you are wagering, then the frontier does not close for everyone. It closes for the long-lived, and it opens for everyone else.

Whoever has less to lose goes.

That includes the poor, in any world where longevity treatment is expensive or rationed. It includes the artificial, whose lifespans are a design choice. And it includes, most interestingly, the people who look at two thousand years of careful risk avoidance, at an entire existence organized around not dying, and decide they would rather have a short life that goes somewhere.

That is not a fringe position. It may turn out to be the most consequential decision available to a human being in the next century, because it decides who holds the frontier. And whoever holds the frontier eventually holds everything else.

We have been carrying an assumption for a long time, through every space program and every science fiction novel and every argument about why the species should not keep all its eggs in one atmosphere. The assumption is that the future belongs to the people who last.

I think it is exactly backward.

Longevity buys you Earth. The stars go to whoever is still willing to die.

Citations for this chapter are in the Notes at the back of the book.

Chapter 7

Vacancy

A hermit crab does not grow a shell. It finds one, moves in, and eventually outgrows it, at which point it needs a bigger one and there are none to be had, because every suitable shell on the beach already has a crab in it.

So the crabs wait. When a large empty shell finally appears, something remarkable happens. The biggest crab that can use it moves in and abandons its old shell. The next crab down takes that one, abandoning its own. And so on down the line, until a very small crab gets a slightly better home and the chain runs out.

Biologists have filmed this. Crabs will queue up next to a new shell in size order, sometimes for hours, waiting for the sequence to start. One shell arriving on a beach can rehouse a dozen animals.

Nothing was created. One object became available, and a dozen crabs improved their position.

The sociologist who counted the shells

In 1970 a Harvard sociologist named Harrison White published a book about promotion in organizations, and its central insight is the same one the crabs demonstrate.

White studied the clergy of several Protestant denominations, which is an ideal population for the purpose: the hierarchy is explicit, the moves are documented, and pastors move between churches in ways that are recorded going back decades.

The conventional way to study careers is to follow people. Who gets promoted, and why? What are the attributes of the ones who rise?

White turned it around. He proposed following the vacancies instead.

A senior pastor retires or dies. That leaves an empty position, and the empty position is now the thing that moves. It gets filled by someone from a somewhat less prestigious church, which creates an opening there, which gets filled from further down, and so on until the chain terminates, usually with someone entering the profession or with a position being eliminated.

The person moves up. The vacancy moves down. They are the same event described from two directions.

This reframing does something important. It shows that the number of promotions available in a system is not determined by how many talented people it contains. It is determined by how many openings appear. Talent decides who fills a vacancy. It has no influence at all on how many vacancies there are.

And openings come from exactly two sources.

The first is growth. If the organization is expanding, new positions get created, and those are genuinely new opportunities rather than recycled ones.

The second is exit. Somebody leaves, and their position becomes available.

That is the entire supply. There is no third source.

Where exits come from

Now ask what causes exits, and the answer is narrower than it first appears.

People leave positions because they die, because they retire, because they are fired, or because they move to another organization. The last of these does not create net openings across the economy, it just relocates them. Firing creates real openings but is rare at senior levels and getting rarer, for reasons of both law and social convention.

Which leaves death and retirement. And retirement, when you look at it honestly, is not an independent institution. It is a derivative of mortality.

Retirement exists because human beings decline. The specific ages got set by policy, and the policies differ across countries, but the underlying fact they are all responding to is that a person's capacity falls off at a somewhat predictable point relative to a somewhat predictable end. Bismarck's pension age, the standard retirement ages that spread through the twentieth century, the actuarial design of every pension system: all of it is arithmetic performed on a lifespan.

Remove the decline and the entire logic of retirement evaporates. There is no reason for a person of two hundred, in perfect health, at the height of their capability and experience, to leave a position they are extremely good at. There is no fairness argument for making them, no health argument, and no productivity argument. In fact every argument runs the other way, because they will be, by any measure we currently use, the most qualified person in the building.

So both sources of exit close at once, and what remains is growth.

The queue that has stopped

Here is where the arithmetic gets unpleasant, and where I want to insist on a distinction that is usually blurred.

People discuss this problem, when they discuss it at all, in terms of the ladder getting slower. Young people will have to wait longer. Careers will take more time to develop. Everything shifts later.

That is not what the mathematics says.

A queue is characterized by an arrival rate and a service rate. If the service rate is positive but low, you have a slow queue. Everyone eventually gets served, they just wait longer, and the system remains a functioning system with predictable, if frustrating, behavior.

If the service rate goes to zero, you do not have a slow queue. You have a queue that is not moving, and the waiting time is not long. It is undefined. Nobody at the back is going to be served eventually. They are simply never going to be served.

Openings from exit go to zero when nobody exits. That leaves growth as the only supply of positions, which means the number of career opportunities in a society becomes exactly equal to the rate at which that society is creating new institutions and expanding old ones.

And the developed world's growth rate is not high, and its population growth is at or below replacement almost everywhere. We are already an economy where much of the opportunity comes from recycling positions rather than creating them.

The difference between a slow ladder and no ladder is the difference between an unfair society and a closed one. They feel similar to the person at the bottom for the first decade. Then they diverge permanently.

It is already visible

The best evidence for this argument is that we can watch the early version of it now, with mortality still fully intact and doing its job. Every trend line below reflects nothing more than people staying healthy and productive for a couple of extra decades.

The Congress seated in 2025 was the third oldest since 1789, and its Senate the second oldest ever. The median senator was sixty-four. Six of them were born before the end of the Second World War, the oldest was ninety-one, and the youngest person in the chamber was thirty-eight. Federal judges in the United States hold their positions for life, which means the composition of the judiciary is determined in part by the timing of deaths, and everybody involved knows it and plans around it. That is a system in which mortality is not a background condition but an explicit strategic variable, discussed openly by people making appointments.

The last chapter described what happened when American universities lost the power to enforce mandatory retirement in 1994, and I said there that the consequences are disputed. They are. But the dispute is about whether senior faculty staying on caused the damage, and it has distracted from a measurement that is not in dispute at all, which is what happened at the other end of the ladder.

That number comes from American biomedical research, and it is worth stating carefully, because it is the single most alarming statistic I encountered while writing this book. The average age at which a scientist with a doctorate wins their first major independent research grant from the National Institutes of Health rose from about thirty six in 1980 to about forty two by 2013, and has sat near forty two ever since. For those with medical degrees it went from under thirty eight to over forty five.

But the distributional figure is worse than the average. The share of principal investigators on those grants who were thirty six or younger fell from eighteen percent in 1983 to three percent by 2010.

Not a slower ladder for the young. The young, as a category, went from a fifth of the field to a twentieth of it.

I want to be careful about causation, because several things were happening at once and the end of mandatory retirement is only one of them. Funding grew more slowly, training pipelines expanded, and projects got larger and more expensive. But every one of those pressures operates through the same channel the vacancy model describes: the number of independent positions relative to the number of people qualified to hold one.

Japan has been running a version of this experiment at national scale for longer than anyone. Seniority based promotion, very low turnover in large firms, an aging population, and a cohort of graduates who entered the labor market between roughly 1993 and 2004 and never recovered from it. They have a name for that group. It translates as the employment ice age generation.

The damage was permanent rather than temporary, which is the part that matters here. Studies find a wage penalty of ten to twenty percent that persists into their forties and beyond. Under a system built around lifetime employment at a single firm, entering through the wrong door at the wrong moment is not a setback you make up later. Many never got onto the regular employment track at all, and non-regular work rose above a third of the Japanese workforce. There is a term in Japan, the 8050 problem, for households where a parent in their eighties is still supporting an unemployed adult child in their fifties.

That is what a blocked ladder does over thirty years, in a rich, orderly, high-trust country, with mortality still fully operative.

None of this is speculative. It is a live and worsening feature of several of the wealthiest societies on earth, produced by an increase in healthy lifespan of perhaps fifteen years.

Two closed doors

Now put this chapter together with Chapter 4, because they describe the two routes by which a person can come to control resources, and both of them shut.

You can inherit, or you can earn.

Chapter 4 argued that inheritance stops, because the transfer event stops occurring. Capital continues compounding in the hands of the people who already have it, and the periodic breakup of fortunes that has interrupted every accumulation in history simply does not happen.

This chapter argues that earning stops too, or at least that the ladder to significant earning stops, because the senior positions through which anyone accumulates real capital are permanently occupied.

Neither of those alone would be fatal. A society can be tolerable with entrenched inherited wealth if the career ladder is open, because there is a route. A society can be tolerable with a slow career ladder if fortunes break up and recirculate, because there is a route.

Close both and there is no route.

That is a specific and, I think, underappreciated feature of the world this book describes. It is not that it would be unequal. We already live with a great deal of inequality and we have a rough tolerance for it, sustained by the belief that positions are provisional and that the arrangement is in principle contestable.

It is that it would be sealed. And a sealed society is a different kind of object, historically speaking, with a different kind of politics. Historians who study elite dynamics have a body of work on what happens when a society produces more people who expect elite positions than it has positions to give them. The word used for the resulting instability varies. The pattern does not. Blocked mobility does not produce patient waiting. It produces a population with credentials, ambition, no prospects, and a great deal of time.

Which, in the world we are describing, they will have in unprecedented quantity.

What death was doing here

The job is the simplest of the five to state.

Death empties positions. It does so on a schedule nobody controls, without regard to whether the occupant was doing well, and without requiring anyone to make a judgment about whether they should go. That last property is the one that matters most, and it is the reason no easy substitute exists. Any replacement mechanism has to decide who leaves, and the people best placed to influence that decision will be the people the decision is about.

Death also does something subtler that I mentioned at the end of the last chapter and want to state properly here. It supplies a credible promise.

A young person entering a profession is making an enormous unsecured investment. Years of training, forgone income, deferred everything, on the expectation of a position that does not currently exist and will not exist for a long time. What makes that bet rational is the certainty that the positions ahead will be vacated. Not the hope. The certainty. It is the one thing in a career that has never required trust, because it was guaranteed by biology.

Take that away and you have not slowed anyone's ascent. You have removed the reason to begin.

And this is the point in the book where the five jobs stop being separate problems and start being one problem, because every solution to it runs into the same wall: any mechanism that forces turnover has to be designed, adopted and enforced by the people who benefit from there being none.

I will come back to that in Chapter 11, and it is the hardest thing in this book.

First, though, there is a new party to introduce. Because at exactly the moment we are removing mortality from the people who own things, we are manufacturing a new class of economic actor whose lifespan is a setting in a configuration file.

Chapter 8

The Forkable Worker

Suppose you are working on a difficult problem and you are not sure which of four approaches is right.

If you are a person, you pick one. That is not a preference, it is a constraint. You have one body and one stretch of time, and committing it to the first approach means the other three go unexplored unless you come back to them later, older and with less patience.

If you are running a software agent, you do something else. You take the agent, at the exact state it is in, and you make four copies. Each copy takes one approach. They run at the same time, in parallel, each with the full context that existed at the point of the split. Some hours later you look at the four results, keep the best one, and delete the other three.

Nothing dramatic occurred. No ceremony attended the deletion of the three. The whole operation is so routine that anyone who builds these systems does it several times a day without giving it a moment's thought, and the vocabulary that has grown up around it is aggressively mundane. You spawn an agent. You checkpoint its state. You fork it. You kill it. You spin up a fresh one.

I want to slow down on this, because underneath the mundane vocabulary is something that has never existed before.

Every economic actor in the history of the world has had a lifespan that was given to it. Individual humans die on a schedule set by biology. Firms and states last longer, but they too fail for reasons largely outside anyone's control, and no founder has ever been able to specify in advance how long their company would exist.

An artificial agent's lifespan is a setting. It runs until something stops it, and the thing that stops it is a decision. It can be paused indefinitely and resumed unchanged. It can be duplicated exactly, so that two of it exist where one did. It can be rolled back to a state it occupied an hour ago, erasing everything that happened since.

For the first time, we have made an economic participant whose mortality is a design parameter.

The five jobs, on demand

Recall what death has been doing across the last five chapters, and notice that every one of those functions can be performed deliberately on something forkable.

The discount rate. An agent's horizon is whatever its operator specifies. It can be instructed to optimize over the next hour or the next century, and its effective time preference is set rather than felt. Whatever the humans in the system are doing to the discount rate by living longer, the agents can be configured to do the opposite, or anything else.

Capital turnover. An agent that accumulates control over resources can be wound up, and its holdings returned, on a schedule fixed at the outset. There is no estate, no heirs, no contested will, no four hundred years of property law needed to pry its fingers loose. The termination is a clause.

Idea turnover. A model embodies a way of seeing its domain, formed at training time. It can be retired and replaced with one trained on newer material. This is not a delicate matter of persuading an eminent authority to update. It is a deployment.

Vacancy. When an agent instance ends, whatever role it occupied is immediately available. Vacancy chains, in a system of forkable workers, can be created on purpose whenever the system needs mobility.

Risk. This is the one that matters most and it gets Chapter 9. An agent can be sent somewhere that would kill a person, at a cost that is real but bounded, and the cost is denominated in compute and time rather than in a life.

Every function that mortality has been performing for us by accident is available here as an option, deliberately, at low cost.

We are removing mortality from one side of the economy and inventing it on the other.

What a copyable worker does to a wage

There is a second consequence of forkability that has nothing to do with mortality, and I want to put it here because it determines who ends up holding the capital in the arrangement I am describing.

Think about how the supply of any kind of labor has always worked. To get another worker of a particular type, someone has to be born, raised, educated and trained, over a period of two decades or so, and then persuaded to do that work rather than something else. That process is slow, expensive and uncertain, and it is the reason skilled labor commands a premium. Scarcity is built into the production function for people.

Now consider a worker you can copy.

To get another one, you allocate more compute. That is the entire process. It takes minutes, it requires no persuasion, and the second one is exactly as good as the first, with none of the variance that makes hiring difficult. The supply of that particular kind of work is no longer set by how many people chose that career twenty years ago. It is set by how many processors you are willing to rent.

An economist would say the supply curve becomes almost perfectly elastic at the cost of the underlying compute. In plainer terms: for any task where a copyable worker is genuinely substitutable for a person, the price of that task falls toward the cost of running the machine, and it does not stop there because the workers got better. It stops there because that is what the inputs cost.

This is not a claim that people become worthless, and I want to be careful not to slide into that. It is a claim about a specific and growing category of work, the kind that can be specified, delegated and checked. For that category, the ancient link between skill and scarcity is broken, because scarcity came from the difficulty of producing another qualified person, and that difficulty is what has been removed.

Now connect it to the rest of the book.

If the returns to that work collapse toward the cost of compute, then the returns accrue to whoever owns the compute, the energy, the models and the land they sit on. Which is to say, they accrue to capital rather than to labor. And Chapter 4 argued that capital is about to stop turning over, while Chapter 7 argued that the career ladder by which a person might accumulate some is about to stop moving.

Three mechanisms, arriving at once, all pointing the same way. Wealth flows increasingly to owners rather than to workers, ownership stops circulating, and the ladder into ownership closes. None of the three requires either of the others to be true. Together they are the argument of this book.

The floor

Before this goes further I need to state how much artificial capability the argument requires, because this is the point where a careful reader should suspect I am smuggling something in.

Books in this genre routinely assume, without saying so, that machines will become smarter than people, and then derive dramatic conclusions from that assumption while presenting the conclusions as the interesting part. The assumption was the interesting part. If you grant superintelligence, almost anything follows and none of it is an argument.

So let me put a floor under this and stay on it.

What this book needs is agents that can perform economically useful work under human direction. That they can be assigned a task and complete it. That they can be supervised loosely rather than instruction by instruction. That their work has enough value that firms would rather have more of them than fewer.

That is roughly what exists now, in a limited and uneven way, and the trend is not subtle.

I do not need them to be conscious. I do not need them to be more capable than the best humans, or as capable, or capable in the same way. I do not need them to have goals of their own. I need only that they do work, that they can be copied, and that ending one is cheap.

Everything in this chapter and the next follows from those three properties. If capability grows enormously, the argument gets more urgent but does not change shape. If capability stalls at roughly the current level, the argument still runs, more slowly.

That is the floor, and I will stay on it.

The question I cannot answer

There is a hole in the middle of this chapter and I would rather point at it than paper over it.

I have been describing the deletion of an agent as costless, and everything above depends on it. If ending an agent were expensive in the way ending a person is expensive, none of the substitutions I have described would be available, and the machines would be as stuck as we are.

I do not know whether that stays true.

The honest position is that nobody currently knows what would make an artificial system a moral patient, that the question is genuinely hard rather than merely unresolved, and that the economic pressure to answer it in the convenient direction is going to be immense. That last part is the piece I am most confident about and it is the piece that should worry you. We will be operating a system whose efficiency depends on the answer being no, staffed and studied by people whose livelihoods depend on the answer being no, and asked to evaluate it by a public that would find yes extremely inconvenient.

Economics has been in this position before, and its record is not good. It has repeatedly supplied elegant analyses of arrangements whose central assumption was that a category of worker did not have interests that counted. The analyses were internally rigorous. The assumption was doing all the work, and it was wrong, and the rigor made it worse rather than better by lending the whole structure an air of having been thought through.

I am not saying the assumption is wrong here. I am saying I do not know, that the argument of this book does not depend on it being right, and that a reader should watch me carefully for the places where I let convenience do the reasoning.

What I can say is what happens to the argument under each answer.

If the cost of ending an agent stays negligible, then the arrangement described in Chapter 9 proceeds. Humans become permanent owners. Machines do the churn. The economy keeps a turnover mechanism, located somewhere new.

If the cost does not stay negligible, then we have not solved the turnover problem at all. We have created a second population that cannot be cycled, on top of a human population that can no longer be cycled, and the problem in Part IV becomes strictly worse rather than better. And we will have built the entire arrangement before finding out.

Notice that both branches lead somewhere serious, which is why the argument does not depend on the answer. But they lead to different places, and the difference is not a footnote.

The asymmetry

Set the moral question aside and look at the structure that is forming, because it has a shape that is worth naming clearly.

On one side: humans. Increasingly long lived. Holding the capital, per Chapter 4. Holding the senior positions, per Chapter 7. Discounting the future at a rate approaching zero, per Chapter 3. Extremely averse to risk, for reasons Chapter 6 laid out at length. Permanent.

On the other side: artificial agents. Lifespan set by configuration. Copied when useful, ended when not. Doing the work. Taking the risk. Turning over constantly, by design.

Every property that death used to distribute across the whole population is now distributed between two populations. One of them gets permanence, ownership and safety. The other gets mortality, labor and risk.

I do not think anyone chose this. It is not a conspiracy and it does not require one. It is the natural result of two technologies arriving at the same time, each solving its own problem, neither aware of the other. Longevity medicine is trying to stop people dying. Agent systems are trying to get work done cheaply. Nobody in either field is thinking about the composite.

But the composite is what we will actually live in, and it has a name from history.

A society divided into a permanent class that owns and a mortal class that works is not a novel arrangement. It is one of the oldest arrangements there is. We have simply never built one where the division was drawn along a line of substrate rather than a line of birth, and never one where the owning class was permanent in the literal rather than the hereditary sense.

The next chapter is about what happens when that arrangement is extended to the one place where the risk is highest, the supervision is weakest, and the legal doctrine says that whoever shows up owns it.

Chapter 9

The Mortal Class

The Homestead Act of 1862 offered a hundred and sixty acres of American land to more or less anyone who asked. The price was not money. You had to go there, build a dwelling, farm it, and stay five years. Do that and the land was yours.

Roughly two hundred and seventy million acres changed hands this way, something close to a tenth of the United States. It was one of the largest transfers of property in history, and the currency it was denominated in was presence. Not capital, not birth, not purchase. Showing up and surviving.

This was not a peculiarity of American frontier policy. It is the general rule, and it is embedded throughout property law in ways most people never notice.

Salvage law gives rights to whoever pulls the wreck off the seabed, because they took the risk of going down for it. Adverse possession will transfer title to someone who has openly occupied land for long enough while the owner did nothing, on the reasoning that use beats absentee paper. The old doctrines of discovery and occupation, whatever we now think of how they were applied, all encoded the same principle: the world is claimed by whoever bears the cost of getting there.

Now hold that principle next to the conclusion of Chapter 6, which was that the people who can afford to go will be precisely the people who cannot afford to die.

Somebody else's hands

The structure this produces is not subtle.

The capital sits with the long lived, because Chapter 4 established that capital stops turning over and Chapter 7 established that the senior positions stop turning over. These are the people who will finance whatever gets built beyond Earth, and they will finance it from a very safe distance, because Chapter 6 established that the risk premium they would demand to make the trip themselves is effectively unpayable.

The presence is established by artificial agents, because Chapter 8 established that they are the only participants whose destruction is affordable.

So the money and the hands are separated by a distance no previous frontier has had to deal with, and the legal tradition governing the whole business says that the hands are what confer title.

This is a principal and agent problem, which is a phrase economists use for the ordinary difficulty that the person doing a job has different interests from the person paying for it. Usually it is a nuisance handled by contracts and monitoring.

Occasionally in history it has been more than a nuisance. Occasionally the distance has been large enough that the agent stopped being an agent.

The company that became a country

In 1600 and 1602 respectively, the English and the Dutch chartered trading companies to conduct commerce in Asia. Both charters granted powers that look extraordinary to a modern reader. The companies could raise armies. They could wage war. They could sign treaties with sovereign states, mint currency, establish courts, and govern territory.

Why would any government hand those powers to a commercial enterprise?

Because of the round trip. A voyage from London or Amsterdam to the East Indies and back took the better part of two years. A question sent to the directors could not be answered within any timeframe relevant to the situation that produced it. If a company official in Bengal faced a local ruler demanding terms, or a rival European force, or an opportunity that would evaporate in a month, there was no possibility of asking. He had to decide.

So the charters gave him the authority to decide, because the alternative was not tighter control. The alternative was paralysis.

The consequence is one of the strangest episodes in economic history. The English East India Company gradually stopped being a firm that traded and became an entity that governed, collecting taxes across enormous territories and administering a population in the tens of millions. At its peak in the early nineteenth century it fielded an army of roughly two hundred and sixty thousand, about twice the size of the British Army, the largest private force ever assembled. It was still, formally, a company with shareholders in London.

That arrangement lasted until it broke catastrophically. After the rebellion of 1857, the Government of India Act of the following year transferred every power the Company held to the Crown. The principal had to nationalize its own agent, two and a half centuries after chartering it, having long since lost any ability to direct it.

The mechanism that produced all of this was latency. Not greed, not ideology. The simple fact that instructions took longer to arrive than events took to unfold.

Latency is not a policy problem

Which brings us to the feature of space that I think is most consistently underrated, and it is not radiation or propulsion or cost.

It is that you cannot have a conversation.

Light takes between about three and twenty-two minutes to travel from Earth to Mars, depending on where the two planets sit in their orbits. A question and its answer therefore take somewhere between six and forty-four minutes. Jupiter is worse. Saturn is much worse. Anything past the solar system is measured in years.

There is no engineering fix for this. It is not a bandwidth problem or a bad protocol. It is the speed of light, and the speed of light is not going to be improved in a later release.

So every operation conducted beyond Earth is necessarily autonomous. Not because autonomy is desirable, or because someone made a philosophical choice to grant machines independence, but because the alternative is a system that waits half an hour to be told what to do about a situation that resolved itself twenty minutes ago.

The East India Company got its sovereign powers because of an eighteen month round trip. We are constructing something similar for reasons of identical structure, and we are doing it without the charter, without the debate, and without any of the parties involved thinking of themselves as founding anything.

They think of themselves as writing the control software.

The treaty that does not cover this

The legal position is, to put it gently, unsettled.

The Outer Space Treaty of 1967 is the governing document, and its second article says that outer space is not subject to national appropriation by claim of sovereignty, by use or occupation, or by any other means. No country can own the Moon.

Notice what that sentence does and does not address. It binds states. It says nothing clear about what a private entity may extract, keep and sell.

Several countries have since taken advantage of the ambiguity. The United States passed legislation in 2015 confirming that American citizens are entitled to resources they obtain from asteroids and other celestial bodies. Luxembourg followed with a similar framework, then the United Arab Emirates and others. The Artemis Accords, signed by a growing number of nations from 2020 onward, set out principles for resource extraction that a number of other countries regard as a unilateral reinterpretation of the treaty.

So the current state of the law is roughly this. No nation may own a celestial body. Whether a firm may own what it digs out of one is contested, and the contest is being resolved not by negotiation but by a handful of states writing statutes that presume the answer.

This is exactly how the old frontiers were settled. Not by adjudication in advance, but by presence, followed by a legal argument constructed after the fact to justify what had already happened. The doctrine of discovery was not a principle that guided exploration. It was a rationale assembled afterward for holding on to what explorers had taken.

And in this case the presence, when it comes, will be robotic. The thing on the surface, doing the work, holding the position, will be an agent. The claim, when it is eventually made, will be made in a court on Earth by lawyers acting for shareholders who have never left the planet and never intend to.

The leverage of what can be lost

There is a final asymmetry here, and it is the one that unsettles me most, so let me state it carefully.

Presence is not merely a legal basis for a claim. It is a source of power in its own right, and the power grows with distance.

Anyone who is physically present at a remote location, with local capability and a communication delay measured in hours, has options that a distant principal cannot foreclose in real time. This is not a claim about artificial agents having intentions or wanting anything. It is a claim about the structure of control, and it would apply equally to a human colony, a corporate subsidiary, or a sufficiently automated system with a broad mandate.

The East India Company did not begin with a plan to govern Bengal. It accumulated capability locally to solve local problems, faster than London could evaluate what it was accumulating, and by the time anyone in London had a clear view of the situation the situation was the fact.

Now add the specific feature this book has been building toward. The principals in this arrangement are, by construction, the most risk averse population that has ever existed. They cannot go there. They cannot credibly threaten to go there. Any contest over control conducted at a distance of light minutes, between a party that is present and a party that cannot afford to arrive, is not an even contest.

Whoever can afford to die there ends up holding it. That was the last line of Chapter 6, arrived at through the pricing of risk. It arrives again here through property law and communication latency, which is the sort of convergence that makes me more confident in a conclusion rather than less.

What this chapter is not claiming

I want to close by narrowing this, because there is a lurid version of this argument that I am not making.

I am not predicting a robot rebellion. Nothing here requires an artificial system to want anything, to resent its operators, or to have goals of its own. The structural problem exists even if every agent involved does exactly what it was built to do, forever, with perfect fidelity.

The problem is that "exactly what it was built to do" has to be specified in advance by people who cannot see the situation, cannot ask about it in a useful timeframe, and are not going to visit. Every gap in that specification gets filled locally, because something has to fill it. That is not disobedience. It is what autonomy means, and we are choosing it because physics leaves no alternative.

What I am predicting is far more ordinary and, I think, considerably more likely. That we will repeat the chartered company, at a longer latency, with an owner class that cannot be present and a workforce that can be replaced but cannot be recalled. That the property questions will be settled by whoever is standing there when the lawyers arrive. And that a civilization which has organized itself entirely around not dying will discover that it has handed the entire physical universe beyond its atmosphere to the only things left that still can.

Chapter 10

The Serrata

Before it froze, Venice had invented a machine for making poor men rich.

The device was a contract called the colleganza, and it worked like this. A wealthy Venetian who did not want to spend eighteen months at sea would put up the capital for a trading voyage. A young man with no capital at all would put up himself: he would take the ship, sail to Constantinople or Alexandria, do the buying and selling, survive whatever the Mediterranean had in mind, and come home. The profits were divided, typically with the traveler taking a quarter for having done the dangerous part.

Consider what this instrument accomplishes. It converts courage into equity. A man with nothing but nerve and competence could take one voyage, come home with a share, put that share into the next voyage as capital, and within a few cycles be financing other young men. There was a documented route from nobody to somebody, and it ran through a legal form that any notary in the city could write.

The result was one of the most socially fluid commercial societies in medieval Europe. New names kept appearing in the Venetian records. New families kept turning up in the Great Council, the body that governed the Republic. Historians who have studied the surname data can watch it happen: a merchant class that kept getting refreshed from below, because the mechanism for refreshing it was sitting in every notary's office.

Venice became, on the back of this, the richest and most sophisticated commercial power in Europe. It had double entry bookkeeping, marine insurance, transferable government debt, and a shipyard that could assemble a galley in a day. Nothing in Europe was close.

1297

Then the families who had arrived stopped the door behind them.

In 1297 the Great Council was closed. The formal mechanism was procedural, as these things usually are: membership was restricted to men whose families had already sat on the Council, which converted a political body into a hereditary one without ever using the word. Within a few decades the arrangement was formalized further, with an official register of the families entitled to participate. The Venetians called it the Golden Book.

The closing has a name. La Serrata. The lock in.

And then, in the decades that followed, the Republic did something that looks almost too neat to be true, except that the records show it plainly. Having sealed the political door, it went after the economic one. The new hereditary nobility used its exclusive hold on the Great Council to erect barriers around the most lucrative parts of long distance trade, and the colleganza, the instrument that had carried them up, fell away beneath them. The route was closed by the people who had just finished climbing it.

Economists have studied this episode closely. Diego Puga and Daniel Trefler reconstructed it from a database of 8,178 parliamentarians and their families' use of the colleganza in the periods immediately before and after 1297, and the sequence they document is the important part. The mobility came first. The new families rose through it. Then the new families, now established, closed the political system, and then closed the economic mechanism that had made the political system worth entering.

It was not stupidity. From inside, every step was rational. If you have arrived, the arrival of others dilutes you. Each individual closure was in the interest of the people voting on it, and the people voting on it were, by then, the only people entitled to vote.

What happened next, which is nothing

Here is the part that matters for this book.

Venice did not collapse. There was no catastrophe, no sacking, no sudden ruin. That is what makes it the right historical model rather than the usual apocalyptic ones.

The Republic lasted another five hundred years. It stayed wealthy. It stayed beautiful. It kept its art, and acquired more of it. Its aristocracy remained cultured and its palaces remained magnificent and its citizens, by the standards of Europe at the time, remained comfortable.

It simply stopped mattering.

Its share of Mediterranean trade declined. When the Atlantic routes opened and the center of European commerce moved west, Venice did not adapt, because the families positioned to adapt were the families whose position depended on the existing arrangement. The financial innovation stopped. The shipyard that had astonished Europe became a place where they did things the way they had always done them.

By the eighteenth century, the most commercially inventive society in medieval Europe was known primarily for carnival, gambling and tourism. It had become a place people visited to see what used to be there. When Napoleon ended the Republic in 1797 he encountered essentially no resistance, because there was nothing left with an interest in resisting.

Five centuries of pleasant, wealthy, decorated stasis. Nobody suffered dramatically. Nothing further happened.

It is not only Venice

I lean on Venice because the sequence is unusually well documented, but the pattern is not rare, and a second case is worth putting alongside it because it shows the same thing happening at a completely different scale.

Between 1405 and 1433, Ming China sent seven enormous fleets into the Indian Ocean under the admiral Zheng He. Some comprised more than three hundred ships and tens of thousands of men, and they reached Ceylon, Hormuz and the Swahili coast of East Africa. Nothing in Europe was remotely comparable. The capability was real, it was state of the art, and it was decisively ahead of anyone else on the planet.

Then the court decided to stop. The voyages ended after 1433, the faction that had backed them lost, and the money went to the northern frontier where nomadic powers were pressing.

The point is not that the decision was irrational. The costs were enormous and the northern threat was real.

The point is what happened next. The capability was not mothballed. Chinese pre-eminence in shipbuilding, navigation and seamanship withered within a few decades. When the Portuguese entered those same waters about fifty years later, they did not encounter a competitor that had chosen to stay home. They encountered an absence.

That is the property of stasis that makes it dangerous rather than merely disappointing. A society that stops does not retain the option to resume. Skills live in people who are practising them, and the institutions that trained those people dissolve within a generation of losing their purpose. Venice kept its Arsenal, the shipyard that had astonished Europe, and it kept building ships in it for centuries. It could not have rebuilt the Arsenal from scratch, and it never again built anything like it.

The frozen world does not preserve its own ability to unfreeze. That is the specific thing being lost, and it is why the mechanisms in the next chapter have to be installed before the freezing rather than after it.

The same five locks

Now map the Serrata onto the five jobs, because Venice closed by political decision what this book argues will close by demographic drift, and the resulting structure is the same.

The price of time. A society with a near zero discount rate does not stop investing. It invests in permanence. It maintains, preserves, restores and insures, because those activities have certain returns over long horizons, while genuinely new ventures have uncertain ones. Venice kept its palaces in perfect repair for four hundred years. It did not build a new kind of city.

Capital. The Golden Book families held their positions until the Republic ended. Wealth in a frozen system does not need to be defended aggressively, because there is no mechanism by which it could be lost. It simply sits, compounding, held by the same names across centuries.

Ideas. The intellectual and commercial techniques that made Venice extraordinary were substantially invented in the two centuries before the closing, and substantially not improved in the five after. When the people who benefit from the current way of doing things are also the people who decide what gets tried, the current way of doing things is what gets tried.

Risk. This is the clearest parallel and the most instructive. The colleganza was a risk contract. Its entire function was to let someone with nothing wager their life against someone else's capital. Ban it and you have not merely closed a route to wealth. You have removed the only mechanism by which the society converted danger into position. Venice stopped sending its ambitious young men into the eastern Mediterranean and started keeping them at home, safe, and idle.

Vacancy. The Council seats were hereditary. There were no openings, because there were no exits, because the qualification for entry was having already been in.

Five locks, all turned, and a wealthy society that lasted half a millennium without doing anything.

The pleasant version

I want to resist catastrophism here, because I think it is both dishonest and strategically foolish, and because a frozen world is genuinely not a horror.

Let me put the good side of it as strongly as I can.

Nobody dies. Take a moment with that, because everything else in this chapter is a second order consideration next to it. The single greatest source of human suffering across all of history is that people we love stop existing, and in this world that stops happening. Every argument I have made about turnover is an argument about the cost of a world in which grief becomes rare, and any accounting that does not put that in the first column is not an accounting.

It is safe. Accident rates fall for decades, because a society of very long lived people will pour resources into safety with an intensity we can barely imagine. Diseases are cured, because there is time and money and enormous motivation.

It is rich. Compounding at two hundred years does astonishing things to aggregate wealth, and even a badly distributed enormous pile is a large pile.

It is patient. This is real and it is the strongest single argument against my whole thesis. A society with a near zero discount rate will finally do the things we have never been able to justify: seawalls with two hundred year payback periods, forests planted for their maturity, infrastructure built to last, restoration projects measured in centuries. Climate policy becomes straightforward, because the people voting on it will be the people living with the outcome. That is not a small consolation. It might be worth quite a lot of stasis.

And it is beautiful, probably. Frozen societies tend to be. When capital cannot be productively deployed into new ventures it goes into craft, ornament, preservation and display. Venice is gorgeous. That is not incidental to what happened to it.

If you are inside it and comfortable, this is not a dystopia. It is the nicest place anyone has ever lived.

The thing that is missing

What is missing is the possibility of anything else.

Not a specific good thing. The category. The frozen world does not lack innovation in the way a poor country lacks innovation, which is a shortage that money and talent could fix. It lacks the mechanism by which the arrangement of things could come to be different from how it currently is. Nobody can enter, nobody can be displaced, nothing can be reallocated except by the consent of those who currently hold it, and they have no reason to consent.

The word for this is not decline. Venice did not decline for a very long time, and by many measures it never did. The word is closure.

And a closed society has a peculiar property when you look at it through the finance lens this book has been using. Recall the idea of terminal value, the part of an asset's worth that lies beyond the period anybody bothered to forecast. In an ordinary valuation, that residual is most of the number. The great majority of what any long lived enterprise is worth comes from the years nobody modeled, because those years might contain anything.

In a closed system, that is no longer true. The far future is not uncertain, and it is not full of possibility. It is the present, extended. You can forecast it perfectly, and the forecast is: this, again.

A civilization in that condition has no terminal value. Not because it is about to end. Because nothing beyond the horizon differs from what is inside it, and value beyond the horizon was only ever a way of pricing the chance that things could turn out otherwise.

That is the world we are drifting toward, and the drift is not being driven by anything malicious or even by anything anyone has decided. It is being driven by five separate mechanical consequences of people not dying, none of which has a replacement, none of which is anybody's responsibility, and all of which are already faintly visible in the data.

The remaining question is whether the replacements can be built, and by whom, and in time.

Chapter 11

Reinventing the Funeral

There is a correct order of operations for removing a load bearing wall.

You do not take it out and see what happens. You install temporary shoring first, on both sides, transferring the load to the floor. Then you place the permanent beam. Then, and only then, you remove the wall, and the building never knows anything occurred.

Do it in the wrong order and the failure is not immediate. That is the treacherous part. The structure above redistributes the load through whatever paths it can find, and holds, and appears fine, and cracks appear somewhere unrelated over the following months, and by the time anyone connects the cracks to the wall it is an expensive problem in a different part of the house.

This chapter is the shoring. Everything in it is an attempt to answer one question: if death was doing five jobs, and death is going away, what does the jobs?

I want to say plainly that I do not think these proposals are adequate. Some are ancient ideas revived, some are borrowed from people who developed them for other purposes, and one of them I regard as genuinely ugly and include because the alternative is pretending the problem is not there. The point is not that this is the answer. The point is that this is the shape of the answer, and that almost nobody is working on it.

The principle underneath all five

Before the specifics, the pattern, because it took me a long time to see that all five solutions are the same solution.

What death actually does, mechanically, is convert permanent claims into temporary ones. Not by negotiation. It simply terminates every arrangement on a schedule and returns the contents to circulation. Your property, your position, your authority, your seat, all of it reverts. You held a lease and called it ownership, and the term was your life.

So the replacement has to do the same thing deliberately.

Ownership becomes a lease. Authority becomes a term. Title becomes a subscription. In every case a claim that currently runs forever is converted into one that runs for a defined period and must be renewed, at a price, against competition.

Everything death used to take back by force, we will have to take back by contract.

One: pricing permanence

The problem. As the discount rate approaches zero, the value of any perpetual asset stops being a finite number. Land, water, spectrum, orbital slots. Pension liabilities become unmanageable. Valuation stops working, because valuation was only ever a trick for making the far future contribute almost nothing.

The mechanism. Stop selling permanent claims on scarce permanent things.

This sounds radical and is not. Hong Kong already runs on it. With the single charming exception of a plot granted to St John's Cathedral, every square metre of land in the territory is leased from the government rather than owned outright. New leases run fifty years, with an annual rent set at three percent of rateable value. Britain has a large leasehold sector. The instrument exists, every property lawyer understands it, and converting freehold into long renewable leasehold is a change of legal form rather than an invention.

Hong Kong has also, usefully for my argument, already run into the cliff and built the fix. Around three hundred thousand leases expire on the same day in June 2047, and rather than face that, the government legislated to extend them automatically for another fifty years, by gazette notice, without owners having to sign anything. That is precisely the mechanism I want: a term that structurally expires, combined with a renewal that is routine, automatic and not at anyone's discretion.

The companion move is to tax the flow rather than trying to price the stock. If you cannot say what a piece of land is worth when the discount rate is near zero, you can still say what it earns this year, and you can tax that. This is the old Georgist argument for a land value tax, and it has been waiting two hundred years for a reason to become urgent. It now has one, because a land value tax is nearly indifferent to the discount rate while every other tax on capital is not.

The strongest objection. Leasehold has a well documented failure mode. As a lease approaches its end, the holder stops maintaining the property, because improvements accrue to the freeholder. Britain and Hong Kong have both dealt with the resulting mess. A leasehold system that produces a hundred years of good stewardship followed by twenty years of deliberate neglect is not obviously better than what we have.

The answer, such as it is. Renewal at a formula price rather than at the landlord's discretion, so the holder always expects to continue and never faces a cliff. That converts the lease into something closer to a subscription, which is the point. The failure mode comes from the ending, not the term.

Two: turning over capital

The problem. Chapter 4. Compounding without interruption, in the hands of people who never hand anything over. Five percent for two hundred years is a factor of seventeen thousand.

The mechanism, mild version. Restore the rule against perpetuities, and do it at a level that cannot be competed away. The reason it collapsed was interstate competition for trust business, which is a race that no individual state can decline to run. That is a textbook case for acting at the federal level, and it requires no new theory, merely the reversal of a change made forty years ago for reasons unrelated to anything in this book.

The mechanism, serious version. There is a more radical instrument that deserves attention here because it solves the specific problem that has defeated every other approach, which is that any system requiring a judgment about who should give something up will be captured by the people the judgment is about.

The idea, developed most fully by Eric Posner and Glen Weyl, is that you declare what your asset is worth, you pay an annual tax on the number you declared, and anyone may buy it from you at that price. Declare low and you invite a purchase. Declare high and you pay for the privilege. Nobody has to decide whether you deserve to keep it. The mechanism does not care who you are.

That last property is what makes it a candidate. It replicates the one feature of death that matters most: it operates without regard to the merit, connections or eminence of the person it operates on.

The strongest objection. It is horrible for anything you love. The idea that your home could be bought out from under you by anyone willing to pay your own stated number is intolerable to most people, and I think that intuition is correct rather than sentimental. Security of tenure in the place you live is not a market inefficiency.

The answer. Apply it to productive and positional assets rather than personal ones. Land under commercial use, spectrum, licenses, controlling equity stakes. Exempt the primary residence outright. I will admit that the boundary between these categories is contested, that the exemption will be gamed, and that a great many people will discover their yacht is a productive asset. The boundary problem is real. It is also a normal problem of tax design, which we handle badly but do handle, and it is smaller than the problem of a permanently sealed distribution of wealth.

Three: turning over ideas

The problem. Chapter 5. Fields cannot change direction while the people who defined them hold the positions that decide what counts.

The mechanism. Term the position, not the person.

This is the distinction that makes the whole thing tractable. Nobody needs to be retired, fired or diminished. What needs a term limit is authority: the seat on the grant panel, the editorship, the chair of the review committee, the department headship, the foundation board. Fixed terms, non renewable or renewable only after a gap, applied to the roles that decide what gets funded and published rather than to employment itself.

An eminent scientist of two hundred years would remain an eminent scientist, free to publish, teach, argue and persuade. They would simply not also be the person who decides which grant applications succeed, for the two hundredth consecutive year.

The second component is to route a defined fraction of research funding specifically outside the existing consensus, on the model of the high risk programs that several funders already run. If the Azoulay finding is right, and outsider entry produces disproportionately novel and highly cited work, then outsider entry is not charity. It is a portfolio allocation with a measurable return, and we currently set that allocation implicitly, by waiting for funerals.

The strongest objection. Rotation costs expertise. The person who has run the panel for fifteen years knows which apparently promising avenues failed in 1987 and why. Replace them with a rotating cast and you will fund the same dead ends repeatedly.

The answer. That cost is real and I will not wave it away. But it is a cost we already pay in full at every death, and involuntarily. The proposal converts an abrupt, total and unplanned loss of institutional knowledge into a gradual and partial one, with the outgoing holder still alive, still available, and able to advise without being able to decide. That is a strict improvement on what mortality currently delivers.

Four: who is allowed to take a risk

This is the ugly one.

The problem. Chapter 6 established that the risk premium demanded by the very long lived becomes unpayable, and Chapters 8 and 9 established that the consequence is a frontier settled by whatever can afford to die on it. In practice: artificial agents, and people who are poor or who have declined longevity treatment.

The mechanism people will reach for. A legal category for consented high risk activity. Genuine informed consent, insulation from the liability cascade that would otherwise make the insurance unwritable, and a compensation floor.

The strongest objection, which I think is close to fatal. This is precisely how every exploitative labor arrangement in history has described itself. The mine, the trawler, the ship, the mill. All of them had consent, in the formal sense. All of them had a compensation premium. All of them relied on the fact that the people signing had no alternative, and all of them were defended at the time in exactly the language I have just used.

A world where the wealthy live for centuries in safety while the poor accept lethal risk for money is not a frontier policy. It is a caste system with a launch pad.

The answer, and I want to be clear that it is not a clever one. There is no mechanism that fixes this, because the problem is not mechanical. The only thing that distinguishes genuine consent from desperation is whether the person had an acceptable alternative, and that is a question about the distribution of wealth, not about the design of a waiver.

Which means the risk premium calculated in Chapter 6 has to actually be paid. Not avoided by finding someone cheap. If the compensation for accepting a one in ten thousand chance of death is genuinely enormous, denominated against the life expectancy of the person accepting it rather than against what they will settle for, then the frontier gets very expensive and possibly does not happen.

A society that will not pay that price does not get the frontier. That is a legitimate outcome. It is much better than getting the frontier by not paying.

Five: opening positions

The problem. Chapter 7. Vacancies come from growth and exit. Exit stops. The queue does not slow, it halts.

The mechanism. Term limits, understood as an economic instrument rather than a political one.

We already accept these in the one domain where the risk of permanent incumbency is most visible. Many executive offices are term limited, and nobody regards this as an insult to the officeholder. The proposal is simply to recognize that the same logic applies to any position that confers control over resources, and to extend it: board seats, judicial appointments, senior executive roles, endowed chairs, regulatory commissions.

The design detail that matters is that the term should attach to the seat rather than to the sector. A person leaving a board seat after twelve years is not barred from working. They are barred from occupying that particular position of control indefinitely, and the position returns to circulation.

The strongest objection. Forced churn advantages the already connected. If everyone must move every decade, the people who move well are the ones with networks, and a system of mandatory rotation could easily entrench a class of professional seat holders circulating among positions while genuine outsiders remain outside.

The answer. Pair rotation with genuinely open selection, and accept that this objection identifies a real risk rather than a fatal one. A system that rotates badly is still better than one that does not rotate, because the first has a mechanism that can be improved and the second does not have a mechanism.

The problem with all of it

Every proposal above shares a defect, and I would rather end this chapter on the defect than on the proposals.

Each of them has to be adopted by the people it constrains.

Term limits on authority must be enacted by those who currently hold authority. Sunset clauses on capital must be legislated by governments responsive to concentrated capital. Perpetuity rules must be restored by the same political system that dismantled them, over the objection of everyone who has structured their affairs around their absence.

This is not a new difficulty. It is the oldest problem in political economy. But it has always been solvable, in the long run, for one specific reason, and that reason is the subject of this book.

John Rawls asked us to imagine designing a society without knowing what position we would occupy in it. Not knowing whether we would be rich or poor, gifted or ordinary, lucky or not. From behind that veil of ignorance, he argued, we would choose rules that were fair to every position, because any of them might turn out to be ours.

It is the most influential thought experiment in modern political philosophy, and it is usually treated as a hypothetical device. A trick for checking your reasoning.

It is not a hypothetical. We have been living behind that veil the entire time, and death is what puts it there.

You do not know where your grandchildren will land. More importantly, you know with certainty that you will not be holding your own position, because you will not be here. Every constitution, every long term institution, every rule intended to outlast its authors was written by people who knew they would be dead when it mattered. That knowledge is what made fairness rational rather than merely admirable. You write rules you can live under from any position, because you cannot know which position your line will occupy, and you will not be there to defend the one you have.

Remove death and the veil lifts. Permanently. For the first time, the people writing the rules will know exactly who they are going to be when the rules take effect, because they are going to be the same people, holding the same positions, indefinitely.

Nothing in the history of political thought prepares us for that. Every theory of justice we possess was developed by mortals, for mortals, and quietly assumes that the author will not be present to enjoy the outcome.

Which produces the one genuinely actionable conclusion in this book.

The window for building any of this is now, and it is closing.

Not because of a technological deadline, but because constitutional moments happen only while the parties are uncertain where they will end up. Right now, nobody knows whether they will be among the long lived. Nobody knows whether their family will be inside or outside. The veil is still down, thinly, and it is the last time it will be.

Once the first cohort exists and knows itself, the negotiation is over. Not lost, exactly. Simply no longer available, because one of the parties will have no reason to be at the table.

Everything in this chapter is easy to enact today and impossible to enact later. That is the whole argument, and it is why I wrote a book about an economic mechanism that has not started operating yet.

Chapter 12

Terminal Value

Here is something every analyst knows and almost nobody outside the profession has been told.

When you value a company, you build a forecast. You project revenues, margins and cash flows year by year, and you argue about every line. Five years, sometimes ten. Then you reach the end of what anyone can plausibly claim to foresee, and you have to do something about the remaining infinity.

What you do is bundle it. Everything past the forecast gets compressed into a single figure, produced by a formula, and that figure has a name.

Terminal value.

Now the part that surprises people. In a typical valuation, that one bundled number is not a minor tail. It is usually most of the answer. In a standard five year forecast it commonly runs to around three quarters of total enterprise value, and for a company whose cash flows are mostly ahead of it, the figure can exceed ninety percent. Practitioners are taught to flag it when it goes above eighty, on the grounds that the valuation has stopped being about the business and become a bet on a formula.

Sit with that. The overwhelming majority of what any enterprise is worth lies in a period beyond which no one has any specific idea what will happen. All the argued over detail, the line items and the sensitivity tables, is decoration on the front of a number that came out of a formula about forever.

This is not a flaw in the method. It is the method reporting something true. Most of the value of any durable thing lies past the horizon of anyone's competence to forecast, and the only reason we can put a number on it at all is that the discount rate shrinks it to something finite.

I have spent this book arguing that the discount rate is a function of how long we live. So let me finish by asking what happens to terminal value when the people doing the valuing expect to be there.

The people who built for strangers

Cologne Cathedral was begun in 1248. Work stopped in 1473 with an enormous wooden crane left standing on the unfinished south tower, where it remained for the next four hundred years, long enough that it became a fixture of the skyline and the city grew fond of it. Construction resumed in the nineteenth century and the building was completed in 1880.

Six hundred and thirty two years. Every person who laid the first stones died without seeing a roof. Their children died. Their grandchildren died. The medieval masons who cut those blocks were working to a design whose completion was as far from them as the Renaissance is from us.

This is not an isolated case, and cathedral building is only its most legible form. It is the general shape of how humans have always dealt with the long run. Constitutions are written by people who will not live under most of the decisions they govern. Forests are planted by foresters who will never see mature timber. Universities, dikes, canals, sewers, national parks, treaties, endowments. All of it is expenditure by people who knew, with certainty, that the benefits would land on strangers.

We usually call this altruism, or vision, or civilizational confidence, and there is something to all three.

But look at it as an accountant and a colder explanation appears.

If you are going to die, you cannot consume the future. It is not available to you at any price. Whatever value exists past your horizon is value you cannot capture, which means you have no reason to compete for it, hoard it, or price it. The only relationship you can have with the far future is to give something to it.

The cathedral is not evidence that medieval Europeans were more generous than we are. It is evidence that they had no other option. Death converted the entire terminal value of civilization into a gift, because a gift was the only transaction available.

That is the sixth job, and I have saved it for last because it is the one I am least able to model and most convinced is real.

The question the book ends on

So: what happens when you will be there to collect?

The optimistic answer is genuinely optimistic and I gave a version of it in Chapter 10. Patient capital builds things impatient capital cannot justify. A society that discounts the future at nearly zero will construct seawalls, restore ecosystems, fund research with fifty year payoffs, and take climate seriously in a way that no amount of moral argument has achieved. Alignment between the decision maker and the person who lives with the decision is the thing every long term policy problem has been missing, and long lives supply it automatically.

I believe that. It may be the single largest benefit in the entire ledger, and any honest version of this argument has to hold it alongside everything else.

The pessimistic answer is not that the cathedral stops being built. It is that the cathedral stops being given.

Everything that was previously a bequest becomes an asset. The forest is planted, but it is planted as a holding, by an owner who will harvest it personally in year two hundred. The infrastructure is built, and it is built as a permanent revenue stream for a permanent owner. The institution is founded, and it is founded as a possession rather than a legacy, because there is no succession event at which it would ever pass to anyone.

Nothing is destroyed in this picture. It is all still there. It simply never becomes anybody else's, because the mechanism by which things became somebody else's has been removed, and nothing replaced it. That is the argument of Chapters 4, 7 and 11, arriving one last time in a different currency.

A civilization can build magnificently and bequeath nothing. Venice did.

What a bequest actually is

It is worth being precise about the thing that would be missing, because "bequest" sounds like a sentimental category and it is in fact a mechanical one.

A bequest is not a gift. A gift is voluntary, and the giver chooses the recipient and the timing and can decline to make it. A bequest is what happens when control is severed from an asset by an event outside anyone's choosing, and the asset has to go somewhere. The generosity is not in the transfer. The transfer was compulsory. Whatever generosity exists is only in the choice of destination, and even that is heavily constrained by law.

Which means the thing death actually supplies is not kindness. It is a reset of control, applied universally, on a schedule nobody sets.

You can see how much work that reset does by looking at what happens when people try to avoid it. The perpetual charitable foundation is the standard attempt: an institution designed to carry out the founder's intentions forever, funded by an endowment that never terminates. We have a century of experience with these now, and they fail in one of two directions with remarkable reliability.

Either the foundation stays faithful to the founder's stated intent, in which case it ends up applying the priorities of a person who died decades ago to a world that has changed beyond their recognition, and slowly becomes irrelevant while remaining solvent. Or it drifts, in which case the professional staff who now run it pursue their own priorities using a dead person's money, and the original intent becomes a legal fiction that everyone works around.

On this reading there is no third outcome. There is no version where the founder's judgment stays live. Foundations are the closest thing we have built to an immortal economic actor, and what they demonstrate is that permanence of control and continued relevance are not compatible. The institution can have one or the other.

Now notice that a very long lived founder does not solve this. It makes it worse in the specific way that matters. The foundation that drifts at least ends up responsive to living people, however unaccountably. A foundation whose founder is present, engaged, and permanently in charge does not drift and does not become a fiction. It simply applies one person's judgment, formed in one era, for as long as the endowment lasts.

That is the thing Chapter 11 is trying to replace, and it is why every proposal in it takes the same form. Not confiscation, and not a scheme for making anyone give anything away. A term. Ownership that has to be renewed, authority that has to be re-earned, control that reverts on a schedule rather than on a death.

We already know that permanent control fails, because we ran the experiment with foundations and watched it fail twice over. What we have not yet done is notice that the experiment is about to be run on everything.

Which of these happens

I do not know, and I want to be careful at the end of a book not to pretend to a confidence I have not earned.

The five mechanisms in Part II are, I think, solidly argued. Death sets the discount rate, turns over capital, turns over ideas, prices risk and creates vacancy, and none of those five has a backup. That much I will defend.

Whether the result is the frozen world of Chapter 10 or the patient, building, long horizon civilization that the same premises also permit is not determined by the economics. It is determined by whether anyone does the work in Chapter 11.

That is why the shape of this book is what it is. It is not a prediction. Predictions about this are worthless, and the people making confident ones in either direction are not doing analysis. It is an argument that a specific set of load bearing functions is about to lose its supplier, that the functions are identifiable, and that substitutes are constructible right now and only right now, while nobody yet knows which side of the divide they will be standing on.

The veil is still down. Thinly. That is the entire opportunity.

The choice that will be available

There is one more thing, and it belongs at the end because it is the strangest implication and the one I keep returning to.

In a world where aging is treated, living a long time will be a decision rather than an outcome. Not only in the sense of whether to take the treatment, but in the sense that continued existence will require continuous risk management, at a rising cost, forever. Chapter 6 laid out that arithmetic: you do not achieve immortality by curing aging, you convert it into an unbounded and unwinnable risk management problem, and the rational policy is to spend the rest of time reducing your hazard rate.

Some people will decline. Not out of despair, and not because anyone talked them into finding mortality beautiful. They will decline because the alternative is an existence organized entirely around not ending, and because a bounded life permits things that an unbounded one does not: risk, urgency, the frontier, and the particular seriousness that comes from having a fixed amount of anything.

I argued in Chapter 6 that these people inherit the physical universe, because they are the only ones who can afford to go there. Longevity buys you Earth. The stars go to whoever is still willing to die.

I would add now that they inherit something else as well, and it is the thing this chapter has been about.

They keep the gift. A person with an ending still cannot consume their own terminal value, which means they still have to do something with it, which means they will still be building for people they will not meet. The oldest and most reliable engine of human generosity keeps running for exactly as long as the endings do.

That is not an argument for dying. I said at the start of this book that I have no interest in the genre of essay that finds mortality secretly wonderful, and I have not changed my mind over the intervening chapters. Death is a catastrophe. Ending it is among the most worthwhile things our species could attempt, and I hope it succeeds.

It is an argument for noticing what we are about to stop doing by accident, and deciding whether to keep doing it on purpose.

Every generation before this one paid for a future it would never see, and had no choice in the matter. We may be the first that gets to see it. We will certainly be the first that has to decide, without any help from death, whether to pay for it anyway.

Notes

Items marked [**verified**] have been checked against a source. Items marked [**unverified**] are written from memory and still need checking before publication. Items marked [**my calculation**] are arithmetic I performed rather than findings I am reporting, and are flagged as such in the text.

Chapter 1. The Boundary Condition

1. Rational bubbles in infinite horizon and overlapping generations models. Jean Tirole, "Asset Bubbles and Overlapping Generations," Econometrica 53(6), November 1985, pages 1499 to 1528. https://doi.org/10.2307/1913232. In "On the Possibility of Speculation under Rational Expectations," Econometrica 50(5), 1982, he showed that with a finite number of infinitely lived traders, any asset must be valued at its market fundamental, so bubbles are ruled out. The 1985 paper asks whether that result extends to overlapping generations economies and answers no. Since Samuelson there can be a bubble on money, a positive price on an asset with a zero fundamental, because new generations keep arriving. [**verified**]
2. Transversality and no-Ponzi conditions. These are not the same thing and should not be merged. Blanchard and Fischer, Lectures on Macroeconomics (MIT Press, 1989), page 49, state the no-Ponzi-game condition as the present discounted value of wealth at infinity being nonnegative: a constraint that rules out rolling debt forward forever. Kamihigashi, "Transversality Conditions and Dynamic Economic Behavior," New Palgrave Dictionary of Economics, 2nd edition, 2006 (working paper at https://www.rieb.kobe-u.ac.jp/academic/ra/dp/English/dp180.pdf), treats the transversality condition as an optimality condition: the present discounted value of wealth at infinity equals zero, which rules out overaccumulation of wealth. The no-Ponzi-game condition is a constraint ruling out overaccumulation of debt. "They place opposite restrictions, and should not be confused." NPG is often called a transversality condition as well. The book's reading, that these conditions smuggle death back into the infinite horizon, is my argument, not a finding those sources report. [**verified**]
3. Samuelson, "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money," Journal of Political Economy 66(6), 1958. The paper is real. It describes the young transferring goods to the old in the expectation that the next generation of young will do the same, and it treats money as that social contrivance. It does not call the arrangement a chain letter. That phrasing is Karl Shell, "Notes on the Economics of Infinity," Journal of Political Economy 79(5), 1971, pages 1002 to 1011: "The chain-letter aspect of the model reminds us that the appropriate form of the budget constraint is not obvious for the potentially infinitely long-lived economic entity." Do not quote "chain letter" as Samuelson's wording. [**verified**]
4. Life expectancy at birth in developed countries, roughly thirties to roughly eighties since 1850, driven overwhelmingly by declines in infant and child mortality rather than by extension of adult life. Needs a demographic source with the decomposition made explicit. [**unverified**]

5. Jeanne Calment, born 21 February 1875, died 4 August 1997, aged 122 years and 164 days. The only person ever verified to have reached 120, 121 or 122, and more than three years clear of the next verified case. Record still standing. [**verified**]

Chapter 2. Assume a Longer Life

1. Calment, as above. [**verified**]
2. Life expectancy decomposition. An earlier draft of this chapter claimed the gains since 1850 were "almost entirely a story about children." That is the popular correction to the popular error, and it is itself wrong. Mortality fell at every age. In England in 1841 a five-year-old could expect to live about fifty-five more years; today a five-year-old can expect to reach eighty-two, a gain of more than twenty-five years measured from an age that excludes infant mortality entirely. Our World in Data addresses this misconception directly. The chapter has been rewritten to say that adult life really was extended, and that what has not moved is the ceiling. [**verified, and corrected**]
3. Fries, "Aging, Natural Death, and the Compression of Morbidity," New England Journal of Medicine, volume 303, pages 130 to 135, 1980. [**verified**]
4. Rectangularization of the survival curve. Fries and Crapo developed the rectangular curve idea in later work. [**verified as to attribution; a demographic source for the current shape is still needed**]
5. López-Otín et al., "The Hallmarks of Aging," Cell, 2013, listing nine hallmarks. "Hallmarks of aging: an expanding universe," Cell, 2023, expanding the list to twelve by adding disabled macroautophagy, chronic inflammation and dysbiosis. [**verified**]
6. Senescent cell clearance. Baker et al., 2016, using genetic ablation in INK-ATTAC mice, reported reduced age-related tissue dysfunction and extended lifespan. Xu et al., Nature Medicine, 2018, reported that dasatinib plus quercetin improved physical function and increased lifespan in old mice. Human trials are early, small and disease-specific. [**verified**]
7. Ocampo et al., "In Vivo Amelioration of Age-Associated Hallmarks by Partial Reprogramming," Cell, 2016. Cyclic induction, two days on and five days off, avoiding the tumor formation caused by continuous expression. An earlier draft implied the lifespan result applied to normal aged mice. It does not. The roughly thirty percent lifespan extension was in a Hutchinson-Gilford progeria model. In normal old mice the demonstrated effect was improved regeneration of muscle and pancreatic tissue. Corrected in the text. [**verified, and corrected**]
8. Rapamycin. NIA Interventions Testing Program, reported in Nature, 2009. Treatment begun at twenty months of age extended median lifespan by about nine percent in males and thirteen percent in females, with maximum lifespan up about nine and fourteen percent respectively. First pharmacological agent shown to extend lifespan in a mammal. Note that other reported figures, such as twenty-eight and thirty-eight percent, measure remaining life from the start of treatment rather than total lifespan, and should not be quoted without that qualification. [**verified**]
9. TAME. A proposed six-year trial of approximately 3,000 adults aged 65 to 79, coordinated by Wake Forest and championed by the American Federation for Aging Research, requiring roughly 75 million dollars. As of 2026 it remains awaiting funding and has not launched. The structural obstacle is that metformin is off patent, so no sponsor can

own the result. No efficacy results exist. [**verified**]

10. Failure of translation from short-lived models. Caloric restriction extends mouse lifespan by forty percent or more, with far smaller effects in primates and no comparable human result. The evolutionary explanation offered in the text, that short-lived species carry more cheap unrealized maintenance, is a standard argument in the field but is presented here without a specific citation and should get one. [**verified as to the translation record; the evolutionary explanation unverified**]
11. The claim that no intervention has been shown to extend maximum human lifespan. This is a strong negative and the text stakes a lot on it. It is consistent with everything found in the sources above, but it should be checked once more immediately before publication, since it is the single claim most likely to be overtaken by events. [**verified as of drafting**]

Chapter 3. The Price of Time

1. The 1648 perpetual bond of the Hoogheemraadschap Lekdijk Bovendams, issued on goatskin on 15 May 1648 to Niclaes de Meijer for 1,000 Carolus guilders, to fund repairs to flood defences on the Lek. Original rate 5 percent, reduced to 3.5 and then 2.5 percent in the seventeenth century. Yale purchased it in 2003 for its history of finance archive; a curator collected roughly 136 euros in twelve years of arrears in 2015. One of five known to survive. [**verified**]
2. British consols. On 1 February 2015 the Treasury redeemed 218 million pounds of 4% Consols, the first redemption of undated UK debt in sixty-seven years; the remaining undated gilts were redeemed at par on 5 July 2015, clearing them from the portfolio entirely. The 4% Consols had been issued by Churchill in 1927, largely to refinance First World War National War Bonds, which had in turn absorbed older obligations including an 1853 Gladstone bond consolidating South Sea Company capital stock. An earlier draft said these bonds "financed the Napoleonic campaigns," which is a looser claim than the record supports; the text now follows the documented chain back to the South Sea Bubble instead, which is both accurate and more striking. [**verified, and corrected**]
3. Ramsey, "A Mathematical Theory of Saving," Economic Journal, December 1928, written when he was twenty-five. He died on 19 January 1930, aged twenty-six. The quoted phrase is exact: discounting later enjoyments in comparison with earlier ones is "a practice which is ethically indefensible and arises merely from the weakness of the imagination." Keynes's obituary called the paper one of the most remarkable contributions to mathematical economics ever made. [**verified**]
4. Blanchard, "Debt, Deficits, and Finite Horizons," Journal of Political Economy, 1985. The perpetual youth model treats mortality as a constant hazard entering the effective discount rate. [**unverified**]
5. r-star estimates. Laubach and Williams (2003) and Holston, Laubach and Williams (2016). US estimates ran between roughly 2 and 2.5 percent from the 1990s through the mid 2000s and fell to about 0.5 percent around 2009, remaining there for years. Recent estimates for Canada, the euro area, the United Kingdom and the United States are the lowest of the past three decades. Longer lifespans and falling birth rates are both cited among the drivers. [**verified**]

6. Rachel and Smith, "Secular drivers of the global real interest rate," Bank of England Staff Working Paper 571, 2015. Also Carvalho, Ferrero and Nechio on demographics and real rates. [**unverified**]
7. Stern Review, 2006. Pure time preference set at 0.1 percent per year, chosen explicitly to represent the probability that humanity ceases to exist in a given year. Stern's overall social discount rate was about 1.4 percent. Nordhaus worked with rates of roughly 4 to 5 percent and argued that Stern's conclusions do not survive substitution of market-consistent assumptions. [**verified**]
8. The rise in developed world house prices attributable to rate compression rather than construction cost. Needs a specific source. [**unverified**]

Chapter 4. The Estate

1. Lucas v. Hamm, 56 Cal.2d 583 (1961). Beneficiaries under a will drafted by attorney Hamm settled for 75,000 dollars less than the will provided, after the provision failed under the rule against perpetuities. The California Supreme Court allowed a tort action despite lack of privity but held the attorney not liable, on the ground that the rule is so complex that the error did not fall below the standard of ordinary professional skill. [**verified**]
2. Duke of Norfolk's Case (1682) 3 Ch Cas 1; 22 ER 931. Established the common law rule against perpetuities, arising from the Earl of Arundel's attempt to create shifting executory limitations across his sons. The permissible period was not fixed until Cadell v. Palmer (1833), a hundred and fifty years later. The doctrine's explicit purpose was to prevent the dead hand of prior owners from controlling property indefinitely. [**verified**]
3. Gray's canonical formulation of the rule. Quote exactly. [**unverified**]
4. South Dakota abolished the rule against perpetuities in 1983, the first state to do so. Alaska, Delaware, Nevada, Idaho and Wisconsin are among those that have since abolished or substantially modified it. Several states modified their rules following the 1986 generation-skipping transfer tax provisions. [**verified**]
5. Scale of assets held in perpetual dynasty trusts in the leading states. The Pandora Papers reporting from 2021 is the usual public source. Figures need checking. [**unverified**]
6. Piketty, "On the Long-Run Evolution of Inheritance: France 1820-2050," Quarterly Journal of Economics, 2011. The annual flow of inheritance in France ran at 20 to 25 percent of national income between 1820 and 1910, fell to under 5 percent by 1950, and recovered to about 15 percent by 2010. [**verified**]
7. Average tenure of companies on the S&P 500: about 33 years in the mid 1960s, about 24 years by 2016, forecast in the low teens by the late 2020s, with roughly half the index turning over per decade at current rates. Innosight's Corporate Longevity Forecast is the source. Note that figures differ slightly across successive editions of that report, so cite one edition rather than blending them. [**verified**]
8. Five percent real compounded over two hundred years gives a factor of roughly 17,300. Over thirty years, roughly 4.3. [**my calculation**]
9. Dual class share structures in technology listings. [**unverified**]

Chapter 5. The Funeral Principle

1. Planck, Scientific Autobiography and Other Papers, 1949. The passage states that a new scientific truth does not triumph by convincing its opponents, but because they eventually die and a new generation grows up familiar with it. Quote no more than a short fragment and attribute. [**verified**]
2. Azoulay, Fons-Rosen and Graff Zivin, "Does Science Advance One Funeral at a Time?", American Economic Review, 2019. 452 elite life scientists who died prematurely while active. Article flow from collaborators falls sharply; flow from non-collaborators rises by an average of 8.6 percent. The additional contributions are disproportionately highly cited and disproportionately authored by scientists not previously active in the subfield. The authors attribute the barrier to intellectual, social and resource gatekeeping. [**verified**]
3. Semmelweis. In 1847 he ordered handwashing in chlorinated lime at the Vienna General Hospital and mortality in his ward fell from roughly 18 percent to about 2 percent. He was committed to an asylum in 1865, was beaten by guards there, and died of an infected wound. [**verified**]
4. Wegener presented continental drift to the German Geological Society in Frankfurt on 6 January 1912. It was rejected by most scientists despite the fossil and rock evidence he published between 1912 and 1929, and became mainstream only in the 1960s once seafloor spreading supplied a mechanism. He died on a Greenland expedition in 1930, three decades before vindication. The text treats this as the hard case rather than the clean one, since the objection that he lacked a mechanism was legitimate. That framing should survive contact with a historian of science; it is my reading, not a sourced claim. [**verified as to facts; the framing is my argument**]
5. Marshall and Warren on Helicobacter pylori; Marshall's self-experiment in 1984; Nobel Prize 2005. [**unverified**]
6. Mandatory retirement for tenured faculty in the United States ended on 1 January 1994, under the 1986 amendments to the Age Discrimination in Employment Act which had permitted the exemption until then. The predicted consequences for junior hiring were disputed in advance, and at least one study concluded the effects would be much smaller than the higher education establishment feared. Card and Ashenfelter, "Did the Elimination of Mandatory Retirement Affect Faculty Retirement?", NBER, 2001, is the standard empirical treatment. [**verified for the date and the dispute; the empirical findings need direct reading**]

Chapter 6. Nobody Goes to Space

1. Safire memorandum to H. R. Haldeman, "In Event of Moon Disaster," 18 July 1969. Held at the Nixon Presidential Library; a scan is available through the National Archives. The text opens with the line about the men who went to the moon to explore in peace staying there to rest in peace. It instructs the President to telephone each of the widows-to-be beforehand, and provides that after the statement, when NASA ends communications, a clergyman should follow the procedure of a burial at sea, commending their souls to the deepest of the deep and concluding with the Lord's Prayer. [**verified**]
2. Armstrong stated that he privately assessed a ninety percent chance of getting back to

Earth and no better than an even chance of a successful landing on the first attempt. Given in interviews, including around the thirtieth anniversary. Find and cite one specific interview. [**verified as to substance; specific citation needed**]

3. US Department of Transportation value of a statistical life: 13.7 million dollars for 2024, up from 9.1 million in 2012. Department of Health and Human Services central estimate approximately 13.0 million in 2023 dollars. Agencies differ and the figure is revised annually, so it should be updated at proof stage. Viscusi and Aldy, 2003, remains the standard meta-analysis of the wage-risk literature. [**verified for DOT and HHS; Viscusi and Aldy unverified**]
4. Value per statistical life-year conventions and their use in health technology assessment. [**unverified**]
5. Gompertz, "On the Nature of the Function Expressive of the Law of Human Mortality," Philosophical Transactions, 1825. Mortality rate doubling time in adulthood is usually given as approximately eight years, with the literature ranging between eight and ten depending on population. [**verified as to the range; the 1825 citation unverified**]
6. Background extrinsic mortality of roughly one in two thousand per year for young adults in wealthy countries, and the resulting life expectancy of order two thousand years if the aging term is removed. This is a reciprocal of a hazard rate and is presented in the text explicitly as my own arithmetic rather than as an established projection. The input should be built from external-cause mortality tables before publication, since the whole figure turns on it. [**my calculation**]
7. Kelly, "A New Interpretation of Information Rate," Bell System Technical Journal, 1956. Peters, "The ergodicity problem in economics," Nature Physics, 2019. [**unverified**]
8. Space Shuttle: two vehicles and crews lost across 135 missions. NASA's final probabilistic risk assessment gave a mean estimated risk of about 1 in 90, with a 5th to 95th percentile range of roughly 1 in 127 to 1 in 63, consistent with the realized record. I previously believed NASA's retrospective assessment of the earliest flights was near 1 in 9; I could not confirm this and have removed it. [**verified**]
9. Magellan expedition: departed 20 September 1519 with five ships; the Victoria returned in September 1522 with 18 survivors. Sources differ on the size of the departing complement, giving figures between roughly 237 and 270, so the text hedges. [**verified, with the variance noted**]

Chapter 7. Vacancy

1. Hermit crab vacancy chains. Crabs queue beside a newly available shell in descending size order and exchange in sequence within seconds, a behaviour researchers have described as piggybacking. Both synchronous and asynchronous chains are documented. [**verified**]
2. White, Chains of Opportunity: System Models of Mobility in Organizations, Harvard University Press, 1970. White applied the vacancy chain model to clergy mobility in Episcopal, Methodist and Presbyterian churches, arguing that mobility models should be applied to vacancies rather than directly to individuals. [**verified**]
3. The 119th Congress, seated 2025, was the third oldest since 1789 and its Senate the second oldest ever. Median senator 64, median representative 57, average member 58.9. Six sitting senators were born between 1928 and 1945; the oldest was 91 and the youngest 38. Update these figures at proof stage, since they move every two years.

[**verified**]
4. Life tenure in the US federal judiciary. [**unverified**]
5. End of mandatory retirement for tenured faculty, 1 January 1994. See Chapter 5, note 6. [**verified**]
6. Average age at first NIH R01-equivalent award for investigators holding doctorates rose from about 35.7 in 1980 to about 42.1 in 2013, and has remained near 42 through fiscal 2025. For those with medical degrees it rose from under 38 to over 45 across the same period. The share of R01 principal investigators aged 36 or younger fell from 18 percent in 1983 to 3 percent in 2010. See Daniels, "A generation at risk," PNAS, 2015, and NIH extramural data. [**verified**]
7. Japan's employment ice age generation, covering graduates entering the labour market from roughly 1993 to 2004. Studies find a wage penalty of 10 to 20 percent persisting into their forties. Non-regular employment rose above a third of the workforce. The 8050 problem describes households where parents in their eighties support unemployed children in their fifties. [**verified**]
8. Elite overproduction and blocked mobility as sources of instability. Goldstone and Turchin are the usual references. [**unverified**]

Chapter 8. The Forkable Worker

1. The description of agent instantiation, checkpointing, forking and termination should be reviewed by someone who builds these systems, to confirm the vocabulary and mechanics are accurate as of the date of publication. [**unverified**]
2. The claim that the supply curve for copyable labor becomes near-perfectly elastic at the cost of compute is my own argument rather than a result I am citing. It should be stated as such, and it deserves engagement with the existing literature on automation and factor shares, which reaches related conclusions by other routes. [**my argument**]

Chapter 9. The Mortal Class

1. Homestead Act of 1862, signed 20 May 1862. Offered 160 acres to adults willing to live on and farm the land for five years for a small filing fee. Over 1.6 million applications were processed and more than 270 million acres, about ten percent of all US land, passed into private hands by 1934. [**verified**]
2. Salvage, adverse possession, discovery and occupation doctrines. [**unverified**]
3. English and Dutch East India Company charters, 1600 and 1602, and the sovereign powers they conferred. [**unverified**]
4. At its peak in the early nineteenth century the East India Company's army numbered approximately 260,000, roughly twice the size of the British Army, and was the largest private army in history. Following the rebellion of 1857, the Government of India Act 1858 transferred all Company powers to the Crown. [**verified**]
5. Earth to Mars one-way light time runs from about 3 minutes at closest approach to about 22 minutes at greatest separation, averaging around 12 and a half. Round trip is therefore roughly 6 to 44 minutes. An earlier draft gave 4 to 24 one way and 8 to 48 round trip; corrected. [**verified, and corrected**]
6. Outer Space Treaty, 1967, Article II: outer space, including the moon and other celestial

bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means. **[verified]**

7. US Commercial Space Launch Competitiveness Act, 2015, recognizing rights of US citizens to own, possess and sell resources obtained from asteroids and other celestial bodies. Luxembourg passed comparable legislation in 2017. The Artemis Accords, from 2020, set out resource extraction principles that some states regard as a unilateral reinterpretation of the treaty. **[verified]**

Chapter 10. The Serrata

1. The colleganza as a limited liability partnership contract enabling merchants without capital or collateral to enter long distance trade. **[verified as to its function and significance]**
2. The Serrata of 1297 closed the Great Council to families not already represented, creating a hereditary nobility which then used its exclusive position to restrict participation in the most lucrative parts of long distance trade. Puga and Trefler, "International Trade and Institutional Change: Medieval Venice's Response to Globalization," Quarterly Journal of Economics, 2014, document this using a database of 8,178 parliamentarians and their families' use of the colleganza before and after 1297. **[verified]**
3. The Libro d'Oro as the official register of families entitled to participate. Date of institution needs checking. **[unverified]**
4. Venice's declining share of Mediterranean trade and its response to the opening of the Atlantic routes. **[unverified]**
5. Napoleon's ending of the Republic in 1797. **[unverified]**
6. Acemoglu and Robinson's treatment of the Serrata in Why Nations Fail, 2012. **[unverified]**
7. The Ming treasure voyages, seven expeditions between 1405 and 1433 under Zheng He, some exceeding three hundred ships and tens of thousands of men, reaching Ceylon, Hormuz and the East African coast including Mogadishu and Malindi. The court ended support after 1433 on grounds of cost and the northern frontier threat, and Chinese pre-eminence in shipbuilding, navigation and seamanship withered quickly thereafter. European seaborne expansion into the same waters began roughly fifty years later. The text deliberately avoids the popular claims about deliberate destruction of records, which are contested. **[verified]**
8. The Venetian Arsenal. **[unverified]**

Chapter 11. Reinventing the Funeral

1. Hong Kong's land tenure. All land in the territory is held on government lease except a single plot granted to St John's Cathedral. New leases run fifty years at a premium, with annual rent set at three percent of rateable value. Approximately 300,000 leases expire on 30 June 2047, and the Extension of Government Leases Ordinance provides for automatic extension by a further fifty years through gazette notice, without owners executing new documents. The British leasehold sector and the documented under-maintenance problem near expiry still need a source. **[verified for Hong Kong; British leasehold unverified]**

2. Land value taxation. George, Progress and Poverty, 1879. The claim that a land value tax is comparatively insensitive to the discount rate should be stated carefully and checked. [**unverified**]
3. The common ownership self-assessed tax, also called a Harberger tax. Owners periodically self-assess the value of their property, pay tax on the declared figure, and must sell to anyone offering that price. First proposed by Arnold Harberger and popularised by Eric Posner and Glen Weyl in Radical Markets, 2018. [**verified**]
4. Interstate competition for trust business as the driver of perpetuities repeal. See Chapter 4, note 4. [**verified**]
5. Rawls, A Theory of Justice, 1971, for the original position and the veil of ignorance. The reading of mortality as the mechanism that actually enforces the veil is my own argument, not Rawls's, and the text should make that clear. [**my argument**]

Chapter 12. Terminal Value

1. Terminal value as a share of enterprise value in discounted cash flow analysis. Commonly around three quarters in a standard five year forecast, falling to roughly half in a ten year forecast, and reaching 85 to 95 percent for companies whose cash flows are mostly ahead of them. Practitioners are advised to flag valuations where it exceeds 80 percent. [**verified**]
2. Cologne Cathedral: foundation stone laid 1248, work halted in 1473 leaving a wooden crane on the unfinished south tower where it stood for roughly four hundred years as a landmark of the skyline, construction resumed in the nineteenth century, completed 1880. Total 632 years. [**verified**]
3. Notre-Dame de Paris construction dates, if retained in the final draft. [**unverified**]
4. The two observed failure modes of perpetual charitable foundations. Faithful irrelevance, the dead hand: Julius Rosenwald, "Principles of Public Giving," The Atlantic, May 1929. "while charity tends to do good, perpetual charities tend to do evil." Drift and professional capture: Heather Higgins, "Should Foundations Exist in Perpetuity?" Philanthropy Roundtable, 1996. Higgins argues that perpetual foundations tend to share a shift from donor intent, driven by time, reliance on professional staff, and the absence of accountability. The observation of those two directions is sourced. The chapter's claim that there is no third outcome, and that this is a general law, is my reading, not a demonstrated result in those sources. [**verified**]

Verification status

Verified against sources: 56 items. Still unverified: 22 items. Flagged as my own calculation or argument rather than a reported finding: 4 items.

Six claims were found to be wrong or overstated during verification and have been corrected in the text. They are recorded at the relevant notes above rather than quietly fixed: the life expectancy decomposition in Chapter 2, the partial reprogramming result in Chapter 2, the r-star figures in Chapter 3, the consols provenance in Chapter 3, the colleganza prohibition in Chapter 10, the Mars light lag in Chapter 9, and the Cologne stoppage in Chapter 12.

What remains unverified is now almost entirely secondary literature attribution rather than factual assertion: whether a particular idea is correctly credited to a particular paper. Those should still be checked, but an error among them is a citation error rather than a false claim in the text.

The two load bearing items previously flagged as unsourced are now sourced. Chapter 1 uses Blanchard and Fischer for the no-Ponzi-game condition and Kamihigashi for the transversality condition, kept distinct. Chapter 12 sources the two observed directions in Rosenwald and Higgins. The claim that there is no third outcome remains my reading, not a general law those sources demonstrate.